Introduction to data tables and data frames using Python (Pandas)

Learning Objectives

By the end of this session, you will be able to:

- Understand **Data Tables and DataFrames** in Python.
- Learn how to create, manipulate, and analyze data using **pandas** and **NumPy**.
- Explore key operations like indexing, filtering, aggregation, and merging.

Different Data Types: Quick Reference

Data Types	Definition
List	An **ordered, mutable collection** of elements that allows duplicates. Defined using [ ].
Tuple	An **ordered, immutable collection** of elements. Defined using ( ).
Dictionary	A **key-value pair collection** that is unordered and mutable. Defined using { }.
Array (NumPy)	A **fixed-type collection of elements** optimized for numerical computations. Requires NumPy.
Series (Pandas)	A **one-dimensional labeled array** where each element has an index. Used in Pandas for data analysis.
DataFrame (Pandas)	A **two-dimensional table-like data structure** with labeled rows and columns. Used for structured data manipulation.

What is a Data Table?

- A Data Table is a structured way to store and organize data in a tabular format with rows and columns.
- It allows for easy access, manipulation, and analysis of data.
- Commonly used in databases (SQL), spreadsheets (Excel), and programming languages (Python Pandas, R, etc.).
- **Used in Python through Pandas DataFrames.**

Example:

ID	Name	Age	Salary
1	Alice	25	50,000
2	Bob	30	60,000
3	Charlie	35	70,000

What is a DataFrame?

- A two-dimensional labeled data structure in Pandas.
- Columns can have different data types (e.g., int, float, string).
- It's similar to a SQL table or a spreadsheet in Excel.

Mathematical Representation:

A DataFrame D with m rows and n columns can be represented as:

$$D = \begin{bmatrix} d_{11} & d_{12} & \dots & d_{1n} \\ d_{21} & d_{22} & \dots & d_{2n} \\ \dots & \dots & \dots & \dots \\ d_{m1} & d_{m2} & \dots & d_{mn} \end{bmatrix}$$

DataFrames in Python

- Creating DataFrames in Python
- Data Operations on DataFrames
- Modifying Data in DataFrames
- Aggregations and Summary Statistics
- Merging and Joining DataFrames
- Handling Missing Values.

Multiple Choice Questions (MCQs)

Q1: What is the primary difference between a Python list and a Pandas DataFrame?

A. Lists store only numbers, DataFrames store only text.

B. Lists are 1D, DataFrames are 2D.

C. Lists are used only in machine learning.

D. DataFrames cannot store numerical data.

Multiple Choice Questions (MCQs)

Q1: What is the primary difference between a Python list and a Pandas DataFrame?

A. Lists store only numbers, DataFrames store only text.

B. Lists are 1D, DataFrames are 2D.

C. Lists are used only in machine learning.

D. DataFrames cannot store numerical data.

Fill in the Blanks

1. A DataFrame is a _____ data structure used for data manipulation in Python.
2. In a Data Table, the horizontal sections are called _____, and the vertical sections are called _____.
3. A dictionary in Python stores data as _____ pairs.
4. A tuple is similar to a list, but it is _____, meaning its values cannot be changed.
5. The process of summarizing and organizing data to extract useful insights is called _____.
6. The vertical sections in a DataFrame, each representing a specific attribute or feature, are called _____.
7. A structured collection of data that is stored electronically and can be queried is known as a _____.
8. The function used to remove missing values from a DataFrame is _____.

Fill in the Blanks

1. A DataFrame is a **two-dimensional** data structure used for data manipulation in Python.
2. In a Data Table, the horizontal sections are called **rows**, and the vertical sections are called **columns**.
3. A dictionary in Python stores data as **key-value** pairs.
4. A tuple is similar to a list, but it is **immutable**, meaning its values cannot be changed.
5. The process of summarizing and organizing data to extract useful insights is called **data analysis**.
6. The vertical sections in a DataFrame, each representing a specific attribute or feature, are called **columns**.
7. A structured collection of data that is stored electronically and can be queried is known as a **database**.
8. The function used to remove missing values from a DataFrame is **dropna()**.

Session Summary

In this session, we have discussed about:

- DataFrames are essential for structured data manipulation.
- Use Pandas for efficient handling of datasets.
- Practice coding problems to master DataFrame operations.

Introduction to Descriptive Statistics

Learning Objectives

By the end of this session, you will be able to:

- Understand **measures of descriptive statistics**: Mean, Median, Mode, Standard Deviation, etc.
- Learn how to **derive and interpret** these measures.
- Use **Python** for hands-on calculation of statistics.

Multiple Choice Questions (MCQs)

What is the primary goal of descriptive statistics?

A) To make inferences about a population

B) To describe and summarize data

C) To test hypotheses

D) To establish causality

Multiple Choice Questions (MCQs)

What is the primary goal of descriptive statistics?

A) To make inferences about a population

B) To describe and summarize data

C) To test hypotheses

D) To establish causality

Answer: B) To describe and summarize data
Descriptive statistics aims to provide a summary of the main features of a dataset, making it easier to understand.

What is Descriptive Statistics?

Descriptive statistics summarize and describe the basic features of a dataset.

Purpose:

- Simplifies complex datasets.
- Identifies patterns, trends, and outliers.
- Forms the foundation for more advanced statistical modeling.

Types of Descriptive Statistics

Measures of Central Tendency: Mean, Median, Mode.

Measures of Dispersion: Range, Variance, Standard Deviation.

Mean

- **Formula:**

$$\text{Mean}(\mu) = \frac{\sum_{i=1}^{n} x_i}{n}$$

Where x_i are the data points and n is the total number of observations.

- **Interpretation:**

Represents the **average** value of a dataset.

$$\text{Mean}(\mu) = \frac{\sum_{i=1}^{n} x_i}{n}$$

Component Breakdown

Symbol	Name	Meaning
$\text{Mean}(\mu)$	**Mean** or **Population Mean**	This is the value you are calculating. The Greek letter μ **(mu)** specifically denotes the **population mean** (the average of *all* possible values). If you were calculating the mean of a **sample** (a subset of the population), the symbol would typically be $\bar{x}$ **(x-bar)**.
$\sum$	**Capital Sigma** (Summation)	This is the **summation** operator. It means you must **add up** everything that follows it.
$i = 1$	**Index Start**	This indicates that the summation starts with the **first** data point in the set (where the index i is 1).
n (above $\sum$)	**Index End**	This indicates that the summation stops at the **last** data point, meaning you sum up to the n-th data point.
x_i	**Data Point**	This represents the individual values in your data set. x_1 is the first value, x_2 is the second, and so on, up to x_n.
$\sum_{i=1}^{n} x_i$	**The Numerator**	This entire expression means: **"Sum up all the data points from the first (x_1) to the last (x_n)."** In simple terms, it's the **total sum** of all the numbers in your set.
n (in the denominator)	**Count**	This is the **total number of data points** in the set.

1. Test Scores

Element (x_i)	Value
x_1	**85**
x_2	**92**
x_3	**78**
x_4	**95**
x_5	**80**

Data Set: $\{85, 92, 78, 95, 80\}$

Step 1: Sum all the values ($\sum x_i$)

We add up all the individual test scores (x_i):

$$\text{Sum} = 85 + 92 + 78 + 95 + 80$$

$$\text{Sum} = 430$$

Step 2: Count the number of values (n)

There are five test scores in the data set:

$$n = 5$$

Step 3: Divide the sum by the count

We substitute the values into the mean formula:

$$\bar{x} = \frac{430}{5}$$

$$\bar{x} = 86.0$$

The arithmetic mean (average) of the test scores is **86.0**.

2. Daily Temperatures (°C)

Element (x_i)	Value
x_1	20
x_2	25
x_3	22
x_4	27
x_5	26

3. Number of Books Read

Element (x_i)	Value
x_1	3
x_2	1
x_3	4
x_4	2
x_5	5

4. Customer Wait Times (Minutes)

Element (x_i)	Value
x_1	4.5
x_2	6.0
x_3	3.5
x_4	5.0
x_5	4.0

5. Number of Cars Sold per Day

Element (x_i)	Value
x_1	10
x_2	10
x_3	10
x_4	10
x_5	10

Median

Definition: The **middle value** when data is sorted in ascending order.

When to Use: Best for **skewed** data or when there are **outliers**.

Example:

*For data [5, 10, 15, 20, 100], the **median** is 15, while the **mean** is skewed by 100.*

1. Test Scores (Odd Count)

Element (x_i)	Value
x_1	85
x_2	92
x_3	78
x_4	95
x_5	80
x_6 (Outlier)	**10**
x_7	88

Data Set: $\{85, 92, 78, 95, 80, \mathbf{10}, 88\}$

Total Count (n): 7

2. Daily Temperatures (°C) (Even Count)

Element (x_i)	Value
x_1	20
x_2	25
x_3	22
x_4	27
x_5	26
x_6	24
x_7 (Outlier)	**45**
x_8	23

Data Set: $\{20, 25, 22, 27, 26, 24, \mathbf{45}, 23\}$

Total Count (n): 8

3. Number of Books Read (Odd Count with a high outlier)

Element (x_i)	Value
x_1	3
x_2	1
x_3	4
x_4	2
x_5	5
x_6	3
x_7 (Outlier)	**30**

Data Set: $\{3, 1, 4, 2, 5, 3, \mathbf{30}\}$

Total Count (n): 7

Mode

Definition: The value that appears **most frequently** in a dataset.

When to Use: Useful for **categorical data** or identifying the most common occurrence.

Example:
If the most common loan amount is $10,000, that's the mode.

Mode Calculation Results

Sample Data Set	Data Set	Frequency Count	Mode
1. Unimodal	$\{5, 8, \mathbf{12}, 5, \mathbf{12}, 15, \mathbf{12}\}$	**12** appears 3 times.	**12**
2. Bimodal	$\{\mathbf{4}, 7, \mathbf{9}, 10, \mathbf{4}, 7, \mathbf{9}, 11\}$	**4** appears 2 times; **9** appears 2 times.	**4 and 9**
3. No Mode	$\{21, 25, 28, 22, 29, 26\}$	All numbers appear 1 time.	**No Mode**
4. Decimals/Currency	$\{1.50, \mathbf{2.25}, 1.50, \mathbf{2.25}, 3.00\}$	**1.50** appears 2 times; **2.25** appears 2 times.	**1.50 and 2.25**
5. Multimodal (w/ Outliers)	$\{\mathbf{100}, 1, \mathbf{100}, \mathbf{200}, 2, \mathbf{200}, \mathbf{100}\}$	**100** appears 3 times; **200** appears 2 times.	**100**

Standard Deviation

Standard deviation is a statistical measure of data dispersion that quantifies how much individual data points differ from the mean (average) of a dataset.

A low standard deviation signifies that data points are clustered closely around the mean, **indicating consistency,** while a **high standard deviation** shows that the data points are spread out over a wider range, **indicating greater variability.**

It is calculated as the square root of the **variance**, which is the average of the squared differences from the mean.

Variance

- **Definition:** The **average of squared differences** from the mean.
- **Formula:**

$$\text{Variance}(\sigma^2) = \frac{\sum_{i=1}^{n}(x_i - \mu)^2}{n}$$

Range and Interquartile Range (IQR)

- **Range:**

$$\text{Range} = \text{Max} - \text{Min}$$

Shows the **spread** between the smallest and largest values.

- **IQR (Interquartile Range):**

Measures the spread of the **middle 50%** of data.

$$\text{IQR} = Q3 - Q1$$

1. Range,
2. IQR (**Interquartile Range**),
3. Variance, and
4. Standard Deviation (SD)

are the four main ways to measure **dispersion** (or spread) in a data set.

Example Data Set: Daily Sales (Units Sold)

We will use the following set of 7 daily sales figures for a small shop.

$$\text{Data Set}(X) = \{12, 18, 20, 25, 25, 30, 42\}$$

Total number of observations (n) = 7.

1. Range

The **Range** is the simplest measure of dispersion. It's the difference between the highest and lowest values in the data set.

$$\text{Range} = \text{Maximum Value} - \text{Minimum Value}$$

- **Maximum Value:** 42
- **Minimum Value:** 12

$$\text{Range} = 42 - 12 = \mathbf{30}$$

The sales figures span a range of 30 units.

Example Data Set: Daily Sales (Units Sold)

We will use the following set of 7 daily sales figures for a small shop.

$$\text{Data Set}(X) = \{12, 18, 20, 25, 25, 30, 42\}$$

Total number of observations (n) = 7.

2. Interquartile Range (IQR)

The **IQR** measures the spread of the **middle 50%** of the data. It is less affected by outliers than the range.

$$\text{IQR} = Q_3 - Q_1$$

Step 2a: Find the Quartiles (Q_1 and Q_3)

The data must first be sorted, which it already is: $\{12, 18, 20, 25, 25, 30, 42\}$.

1. **Find the Median (Q_2):** Since $n = 7$ (odd), the median is the middle value, which is the $\frac{7+1}{2} = 4^{\text{th}}$ position.

$$\{12, 18, 20, \mathbf{25}, 25, 30, 42\} \implies \mathbf{Q_2 = 25}$$

2. **Find the First Quartile (Q_1):** This is the median of the lower half of the data (the values before Q_2): $\{12, 18, 20\}$.

$$\{12, \mathbf{18}, 20\} \implies \mathbf{Q_1 = 18}$$

3. **Find the Third Quartile (Q_3):** This is the median of the upper half of the data (the values after Q_2): $\{25, 30, 42\}$.

$$\{25, \mathbf{30}, 42\} \implies \mathbf{Q_3 = 30}$$

Step 2b: Calculate IQR

$$\text{IQR} = Q_3 - Q_1 = 30 - 18 = \mathbf{12}$$

The middle 50% of daily sales figures has a spread of 12 units.

Example Data Set: Daily Sales (Units Sold)

We will use the following set of 7 daily sales figures for a small shop.

$$\text{Data Set}(X) = \{12, 18, 20, 25, 25, 30, 42\}$$

Total number of observations (n) = 7.

3. Variance (s^2) and 4. Standard Deviation (s)

Variance and Standard Deviation measure the average distance of each value from the mean. We will calculate the **Sample Variance (s^2)** and **Sample Standard Deviation (s)** using the $n-1$ denominator (Degrees of Freedom).

Step 3a: Calculate the Mean ($\bar{x}$)

$$\bar{x} = \frac{\sum x}{n} = \frac{12 + 18 + 20 + 25 + 25 + 30 + 42}{7} = \frac{172}{7} \approx \mathbf{24.57}$$

Step 3b: Calculate Deviations and Squared Deviations

x_i (Value)	$x_i - \bar{x}$ (Deviation)	$(x_i - \bar{x})^2$ (Squared Deviation)
12	$12 - 24.57 = -12.57$	158.00
18	$18 - 24.57 = -6.57$	43.16
20	$20 - 24.57 = -4.57$	20.88
25	$25 - 24.57 = 0.43$	0.18
25	$25 - 24.57 = 0.43$	0.18
30	$30 - 24.57 = 5.43$	29.48
42	$42 - 24.57 = 17.43$	303.80
Sum ($\sum$)	$\approx \mathbf{0}$	$\sum(x_i - \bar{x})^2 \approx 555.68$

Step 3c: Calculate the Sample Variance (s^2)

$$\text{Variance } (s^2) = \frac{\sum(x_i - \bar{x})^2}{n-1}$$

$$\text{Variance } (s^2) = \frac{555.68}{7-1} = \frac{555.68}{6} \approx \mathbf{92.61}$$

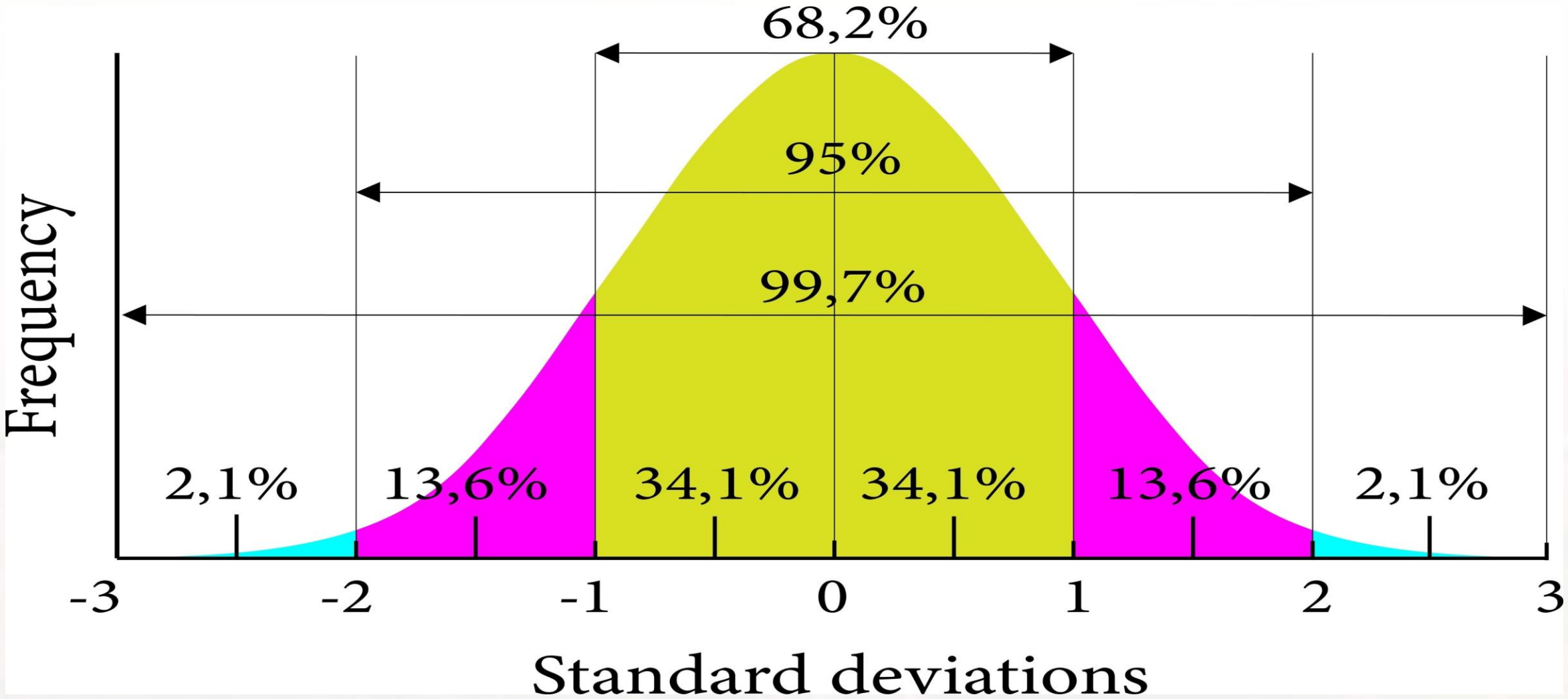
Step 4: Calculate the Sample Standard Deviation (s)

The Standard Deviation is the square root of the Variance.

$$\text{Standard Deviation } (s) = \sqrt{\text{Variance}} = \sqrt{s^2}$$

$$\text{Standard Deviation } (s) = \sqrt{92.61} \approx \mathbf{9.62}$$

This means that, on average, a day's sales deviates from the mean sales figure ($\bar{x} = 24.57$) by about 9.62 units.


68,2%
95%
99,7%
Frequency
2,1%
13,6%
34,1%
34,1%
13,6%
2,1%
-3
-2
-1
0
1
2
3
Standard deviations

Measure of Dispersion	Result	Interpretation
Range	30	The entire data set spans 30 units (from 12 to 42).
IQR	12	The middle 50% of sales are spread over 12 units.
Variance (s^2)	92.61	The average of the squared differences from the mean (units squared, harder to interpret directly).
Standard Deviation (s)	**9.62**	The average distance each sales figure is from the mean ($\bar{x}=24.57$) in the original units.

Questions?

1. Which measure of central tendency is most affected by outliers?

A) Median B) Mode C) Mean D) Range

2. What does a standard deviation of 0 indicate?

A) All values are different B) Data is widely spread C) All values are the same D) There are outliers in the data

3. If the mean and median are equal, the data is likely:

A) Skewed right B) Symmetrical C) Skewed left D) Uniform

Questions?

1. Which measure of central tendency is most affected by outliers?

A) Median B) Mode C) Mean ✅ D) Range

2. What does a standard deviation of 0 indicate?

A) All values are different B) Data is widely spread C) All values are the same ✅ D) There are outliers in the data

3. If the mean and median are equal, the data is likely:

A) Skewed right B) Symmetrical ✅ C) Skewed left D) Uniform

Python Implementation

- Credit Risk Analysis for LendingClub.com
- You are a data scientist working at LendingClub.com and you have been asked to perform descriptive statistics on the data….

Sl.no	Variable Name	Description
1	credit.policy	1 if the customer meets the credit underwriting criteria of LendingClub.com, 0 otherwise.
2	purpose	The purpose of the loan (e.g., "credit_card", "debt_consolidation", "educational", etc.).
3	int.rate	The interest rate of the loan as a proportion (e.g., 11% is stored as 0.11). Higher risk equals higher rates.
4	installment	The monthly installments ($) owed by the borrower if the loan is funded.
5	log.annual.inc	The natural log of the self-reported annual income of the borrower.
6	dti	The debt-to-income ratio of the borrower (debt amount divided by annual income).
7	fico	The FICO credit score of the borrower.
8	days.with.cr.line	The number of days the borrower has had a credit line.
9	revol.bal	The borrower's revolving balance (amount unpaid at the end of the credit card billing cycle).
10	revol.util	The borrower's revolving line utilization rate (credit line used relative to total credit available).
11	inq.last.6mths	The borrower's number of inquiries by creditors in the last 6 months.
12	delinq.2yrs	The number of times the borrower had been 30+ days past due on a payment in the past 2 years.
13	pub.rec	The borrower's number of derogatory public records (bankruptcy filings, tax liens, judgments).
14	not.fully.paid	Binary variable indicating if the loan was not paid back in full (defaulted or charged off).

Session Summary

In this session, we have discussed about:

- **Descriptive statistics** help summarize and interpret datasets.
- **Central tendency** measures (Mean, Median, Mode) describe data location.
- **Dispersion** measures (Standard Deviation, Variance, IQR) show variability.

Reference Links

- Mastering Credit Risk Analysis: A Step-by-Step Guide to Descriptive Statistics in Python
- Top 20 Frequently Asked Questions on Descriptive Statistics for Data Science Aspirants.
- How Much Statistics Do You Really Need for Data Science?
- Phases of data science and analytics.

Introduction to Data Visualization

Why Visualization Skills Matter in Careers?

01

Critical for roles like Data Analyst, BI Analyst, and Data Scientist.

02

Enhances your ability to present data-driven insights.

03

Visualization tools (e.g., Tableau, Power BI) are tested in interviews.

Learning Objectives

By the end of this session, you will be able to:

- Understand **the importance of data visualization.**
- Understand the various **visualization techniques.**
- Learn how to **create and interpret** the various plots in Python.

Data Visualization in Decision-Making



Helps stakeholders understand trends and patterns quickly.



Increases the impact of presentations and reports.



Critical for strategy formulation and operational insights.

How does Data Visualization help?

Transforms raw data into clear, actionable insights.

Enhances decision-making by making patterns easier to detect.

Key for storytelling and communicating findings effectively.

Common Visualization Techniques



Line Charts: Best for tracking data over time (e.g., COVID cases).



Bar Charts: Effective for comparing categories.



Pie Charts: Visualizing part-to-whole relationships.



Scatter Plots: Showing relationships between variables.

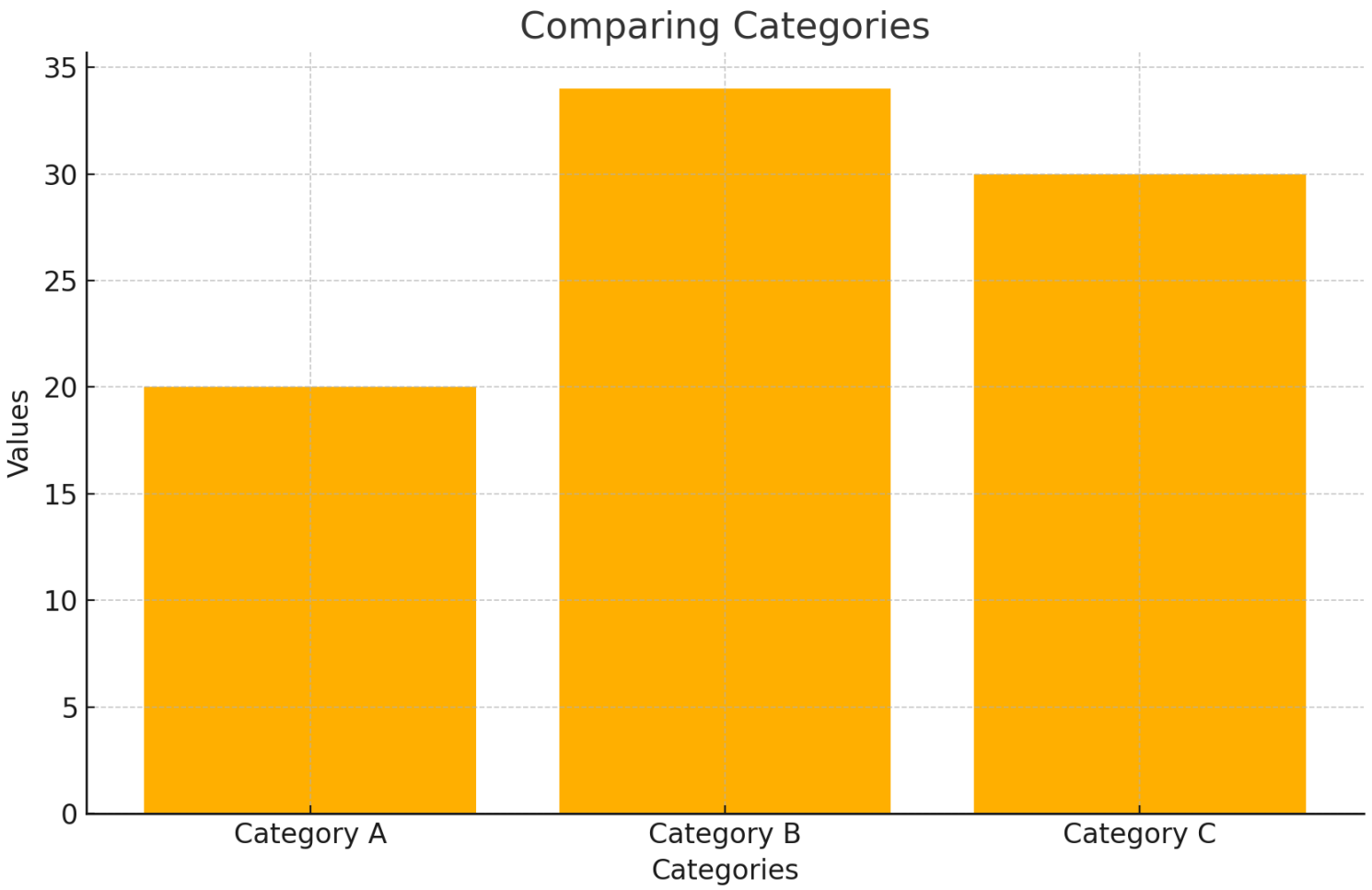


Heat Maps: Visualizing density or intensity of data.

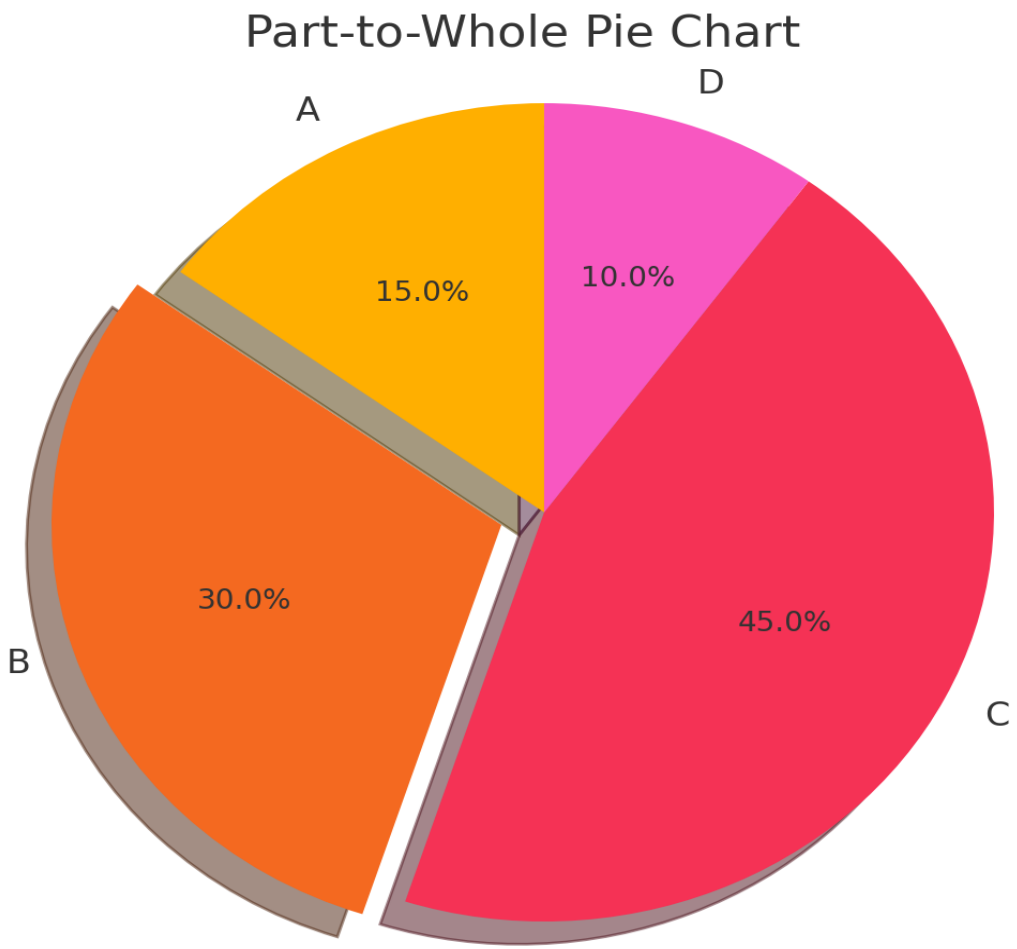
Line Chart Example: Stock Price of Bata India as on Sep 9



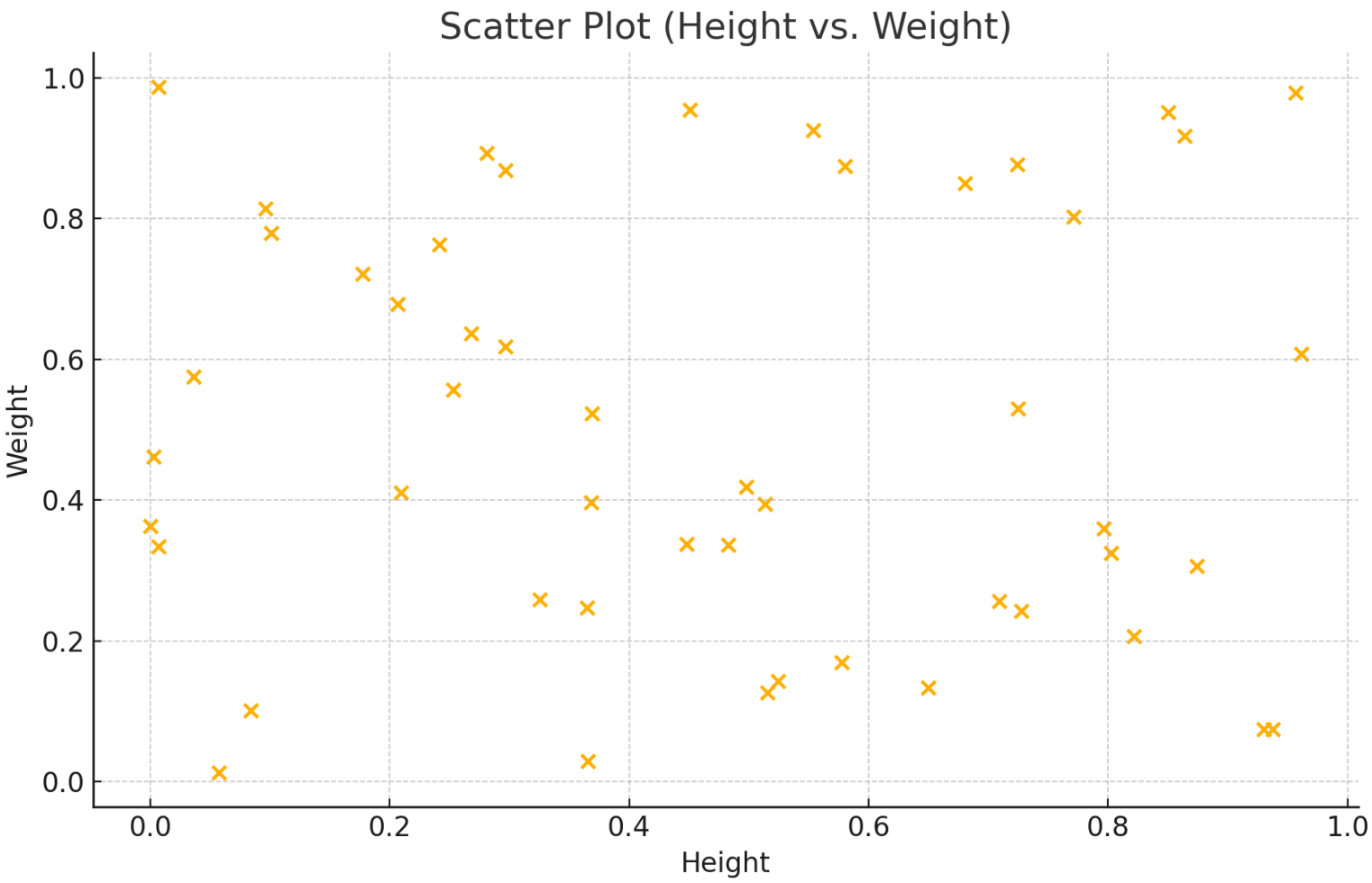
Bar Chart Example: Comparing Categories



Pie Chart Example: Part-to-Whole



Scatter Plot Example: Height vs Weight



How to Choose the Right Visualization

1. Identify the type of data you are working with.
2. Understand the audience and the message you want to convey.
3. Choose charts that simplify complex data.
4. Avoid over-complicating with too many visual elements.

Multiple Choice Questions (MCQs)

1. A data scientist wants to show the monthly trend of website traffic over a year. Which visualization is most appropriate?

A. Scatter Plot
B. Line Chart
C. Pie Chart
D. Bar Plot

2. A researcher wants to examine the relationship between students' study hours and their exam scores. Which visualization should be used?

A. Pie Chart
B. Line Chart
C. Scatter Plot
D. Bar Plot

Sample Questions

Sample Questions:

- "How would you approach visualizing sales performance across regions?“
- "Can you explain a time when a visualization led to actionable business insights?"
- Hint: Practice explaining your thought process in both technical and business terms.

Python Implementation

- ‘mpg’ dataset in seaborn
- You are a data scientist working at car sales outlet and you have been asked to perform EDA using visualization on the data….

	mpg	cylinders	displacement	horsepower	weight	acceleration	model_year	origin	name
0	18.0	8	307.0	130.0	3504	12.0	70	usa	chevrolet chevelle malibu
1	15.0	8	350.0	165.0	3693	11.5	70	usa	buick skylark 320
2	18.0	8	318.0	150.0	3436	11.0	70	usa	plymouth satellite
3	16.0	8	304.0	150.0	3433	12.0	70	usa	amc rebel sst
4	17.0	8	302.0	140.0	3449	10.5	70	usa	ford torino

Session Summary

In this session, we have discussed about:

- The **importance of visualization** from career as well as business perspectives.
- The various **visualization techniques** and when to us each one of these.
- How to **create and interpret** the various plots in Python.

Reference Links

- How to Create Stunning Data Visualizations in Python: Top 10 Techniques to Learn
- Mastering Credit Risk Analysis: A Step-by-Step Guide to Descriptive Statistics in Python
- How Much Statistics Do You Really Need for Data Science?
- Phases of data science and analytics.

Understanding Features

Data Features.....

- ‘titanic’ dataset in seaborn

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

Learning Objectives

By the end of this session, you will be able to:

- Define categorical, discrete, and continuous features.
- Understand the differences and significance of each type.
- Explore examples with real-world datasets.
- Use Python (Pandas, NumPy, Matplotlib, Seaborn) for data exploration.

Types of Features

1

1. Categorical Features: Represent labels or categories (e.g., gender, country).

2

2. Discrete Features: Represent countable values (e.g., number of children).

3

3. Continuous Features: Represent measurable values (e.g., height, salary).

Categorical Features

- Definition: Features that represent categories or groups.

Examples:

- Gender: Male, Female
- Car Type: Sedan, SUV, Truck
- Payment Method: Credit Card, Cash, Digital Wallet

Data Features.....

- Which of these are Categorical features?

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

deck	Categorical	Nominal	Represents the deck letter (A, B, C, etc.). While there might be a physical order, the deck itself is often treated as a set of distinct, unordered groups in analysis unless more context is given.
embark_town	Categorical	Nominal	Full name of the port of embarkation (Southampton, Cherbourg). No inherent order.
embarked	Categorical	Nominal	Represents the port of embarkation (S, C, Q). The ports have no inherent order.
sex	Categorical	Nominal	Represents distinct, unordered groups (male, female).
who	Categorical	Nominal	Represents a category (man, woman, child).
adult_male	Categorical	Nominal (or Binary/Dichotomous)	A logical category (True/False) indicating if the person is an adult male.
alive	Categorical	Nominal (or Binary/Dichotomous)	A category (yes/no) for survival.
alone	Categorical	Nominal (or Binary/Dichotomous)	A logical category (True/False) indicating if the person was alone.
survived	Categorical	Nominal (or Binary/Dichotomous)	Represents a group (survived/not survived). The numbers (0 and 1) are just labels for the categories.
class	Categorical	Ordinal	Text representation of pclass (First, Second, Third), showing a meaningful order.
pclass	Categorical	Ordinal	Represents the passenger class (1st, 2nd, 3rd). The classes have a meaningful, ordered ranking.

1) Categorical Columns (Nominal and Ordinal)

These columns represent qualities or attributes and can be divided into groups.

- **Nominal (Names/Labels with no order):**
  - `survived` (0, 1)
  - `sex` (male, female)
  - `embarked` (S, C, Q)
  - `who` (man, woman)
  - `adult_male` (True, False)
  - `deck` (C, NaN)
  - `embark_town` (Southampton, Cherbourg)
  - `alive` (yes, no)
  - `alone` (True, False)
- **Ordinal (Labels with a meaningful order):**
  - `pclass` (1, 2, 3)
  - `class` (First, Second, Third)

2) Discrete Columns

These columns contain numerical data that are countable and can only take on a finite number of values (typically integers).

- `sibsp`
- `parch`

3) Continuous Columns

These columns contain numerical data that can take any value within a range (often includes fractions or decimals).

- `age`
- `fare`

Types of Categorical Features

1. Nominal: No specific order (e.g., colors: Red, Blue, Green).

2. Ordinal: Has an inherent order (e.g., education levels: High School, Bachelor’s, Master’s).

Discrete Features

- Definition: Numerical features that take countable, finite values.

Examples:

- Number of children in a family
- Number of products purchased
- Number of login attempts

Data Features.....

- Which of these are Discrete features?

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

parch	Discrete	Ratio	Count of parents/children aboard. It must be an integer and the value of 0 is a true zero.
sibsp	Discrete	Ratio	Count of siblings/spouses aboard. It must be an integer and the value of 0 is a true zero.

Mathematical Properties of Discrete Features

Countable values with no intermediate values.

Follows probability mass functions (PMF).

Example: Number of calls received per day:

$$P(X = x) = \frac{\text{Frequency of } x}{\text{Total number of observations}}$$

Continuous Features

- Definition: Numerical features that can take any value within a range.

Examples:

- Height and weight
- Temperature and pressure
- Income and expenses

age	Continuous	Ratio	Can take any value within a given range (e.g., 22.0, 38.0). Zero is an absolute minimum.
fare	Continuous	Ratio	Can take any value within a given range, representing cost.

Mathematical Properties of Continuous Features

Can take infinite values within a range.

Follows probability density functions (PDF).

Example: Height of individuals:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- μ = Mean
- σ = Standard deviation

Python Implementation

- ‘titanic’ dataset in seaborn

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

Multiple Choice Questions (MCQs)

Q1: Which feature type is used to represent customer feedback: “Satisfied,” “Neutral,” “Dissatisfied”?

A) Categorical

B) Discrete

C) Continuous

D) Ordinal

Q2: The number of books a person owns is an example of which feature type?

A) Categorical

B) Discrete

C) Continuous

D) Ordinal

Multiple Choice Questions (MCQs)

Q1: Which feature type is used to represent customer feedback: “Satisfied,” “Neutral,” “Dissatisfied”?

A) Categorical ✅

B) Discrete

C) Continuous

D) Ordinal

Q2: The number of books a person owns is an example of which feature type?

A) Categorical

B) Discrete ✅

C) Continuous

D) Ordinal

Fill in the blanks

1. _____ features represent categories or groups.

2. A _____ feature can take any value within a range.

3. The count of items purchased is an example of a _____ feature.

Fill in the blanks

1. _____ features represent categories or groups. (Answer: Categorical)

2. A _____ feature can take any value within a range. (Answer: Continuous)

3. The count of items purchased is an example of a _____ feature. (Answer: Discrete)

Session Summary

In this session, we have discussed about:

- Define categorical, discrete, and continuous features.
- Understand the differences and significance of each type.
- Explore examples with real-world datasets.
- Use Python (Pandas, NumPy, Matplotlib, Seaborn) for data exploration.

Sample E-commerce Order Dataset

Order_ID	Product_Category	Review_Score	Quantity	Price_Per_Unit	Delivery_Region	Discount_Applied	Customer_Age
1001	Electronics	**5**	1	499.99	North	True	34
1002	Clothing	**3**	2	45.50	South	False	22
1003	Books	**4**	5	12.99	West	False	58
1004	Home Goods	**1**	1	89.00	East	True	41
1005	Electronics	**5**	1	129.99	West	False	29
1006	Clothing	**4**	3	25.00	North	True	67
1007	Books	**2**	10	8.50	South	False	19
1008	Home Goods	**5**	1	350.75	East	False	50
1009	Clothing	**3**	2	65.99	North	False	38
1010	Electronics	**4**	1	199.99	West	True	45

Column	Feature Type	Sub-Type	Reasoning
Order_ID	**Categorical**	Nominal	Unique identifier/label. No inherent order or numerical meaning.
Product_Category	**Categorical**	Nominal	Distinct, unordered groups (Electronics, Clothing, etc.).
Review_Score	**Categorical**	**Ordinal**	Scores (1 to 5) have a meaningful, ordered ranking.
Quantity	**Discrete**	Ratio	Countable, whole number of items sold.
Price_Per_Unit	**Continuous**	Ratio	Can take any value within a range (includes decimals like 499.99).
Delivery_Region	**Categorical**	Nominal	Distinct, unordered geographical labels.
Discount_Applied	**Categorical**	Nominal (Binary)	A yes/no or True/False label.
Customer_Age	**Discrete**	Ratio	Countable, whole number of years (often treated as continuous for modeling purposes, but technically a count).

The Iris dataset contains **150 observations** (rows) of Iris flowers, with **three species** observed. For each flower, it records four features related to the physical dimensions of its petals and sepals, plus the species itself.

Iris dataset

sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
5.1	3.5	1.4	0.2	setosa
4.9	3.0	1.4	0.2	setosa
4.7	3.2	1.3	0.2	setosa
4.6	3.1	1.5	0.2	setosa
5.0	3.6	1.4	0.2	setosa
5.4	3.9	1.7	0.4	setosa
4.6	3.4	1.4	0.3	setosa
5.0	3.4	1.5	0.2	setosa
4.4	2.9	1.4	0.2	setosa
4.9	3.1	1.5	0.1	setosa

Column	Example Values	Feature Type	Sub-Type	Reasoning
sepal_length	5.1, 4.9, 7.0	**Continuous**	Ratio	Can take any value within a range; measured physical dimension.
sepal_width	3.5, 3.0, 3.2	**Continuous**	Ratio	Can take any value within a range; measured physical dimension.
petal_length	1.4, 1.4, 5.1	**Continuous**	Ratio	Can take any value within a range; measured physical dimension.
petal_width	0.2, 0.2, 1.8	**Continuous**	Ratio	Can take any value within a range; measured physical dimension.
species	Setosa, Versicolor, Virginica	**Categorical**	Nominal	Represents distinct, unordered groups of flowers.

Wine Quality Data

fixed_acidity	volatile_acidity	citric_acid	residual_sugar	chlorides	free_sulfur_dioxide	total_sulfur_dioxide	density	pH	sulphates	alcohol	quality
7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978	3.51	0.56	9.4	**5**
7.8	0.88	0.00	2.6	0.098	25.0	67.0	0.9968	3.20	0.68	9.8	**5**
7.8	0.76	0.04	2.3	0.092	15.0	54.0	0.9970	3.26	0.65	9.8	**5**
11.2	0.28	0.56	1.9	0.075	17.0	60.0	0.9980	3.16	0.58	9.8	**6**
7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978	3.51	0.56	9.4	**5**
7.4	0.66	0.00	1.8	0.075	13.0	40.0	0.9978	3.51	0.56	9.4	**5**
7.9	0.60	0.06	1.6	0.069	15.0	59.0	0.9964	3.30	0.46	9.4	**5**
7.3	0.65	0.00	1.2	0.065	15.0	21.0	0.9946	3.39	0.47	10.0	**7**
7.8	0.58	0.02	2.0	0.073	9.0	18.0	0.9968	3.36	0.57	9.5	**7**
7.5	0.50	0.36	6.1	0.071	17.0	102.0	0.9978	3.35	0.80	10.5	**5**

Column	Feature Type	Sub-Type	Reasoning
fixed_acidity	**Continuous**	Ratio	Represents concentration (g/dm^3). It can take any value within a range.
volatile_acidity	**Continuous**	Ratio	Represents concentration (g/dm^3). It can take any value within a range.
citric_acid	**Continuous**	Ratio	Represents concentration (g/dm^3). It can take any value within a range.
residual_sugar	**Continuous**	Ratio	Represents concentration (g/dm^3). It can take any value within a range.
chlorides	**Continuous**	Ratio	Represents concentration (g/dm^3). It can take any value within a range.
free_sulfur_dioxide	**Continuous**	Ratio	Represents concentration (mg/dm^3). It can take any value within a range.
total_sulfur_dioxide	**Continuous**	Ratio	Represents concentration (mg/dm^3). It can take any value within a range.
density	**Continuous**	Ratio	Represents mass per volume (g/cm^3). It can take any value within a range.
pH	**Continuous**	Interval	A measure of acidity/basicity. Zero is arbitrary (it doesn't mean zero acidity).
sulphates	**Continuous**	Ratio	Represents concentration (g/dm^3). It can take any value within a range.
alcohol	**Continuous**	Ratio	Represents percentage of alcohol by volume. It can take any value within a range.
quality	**Categorical**	**Ordinal**	Represents a score (typically 0-10) where higher values indicate better quality, establishing a meaningful order.

Breast Cancer Diagnosis Data

mean radius	mean texture	mean perimeter	mean area	mean smoothness	diagnosis
17.99	10.38	122.80	1001.0	0.11840	malignant
20.57	17.77	132.90	1326.0	0.08474	malignant
19.69	21.25	130.00	1203.0	0.10960	malignant
11.42	20.38	77.58	386.1	0.14250	malignant
20.29	14.34	135.10	1297.0	0.10030	malignant
12.45	15.70	82.57	477.1	0.12780	malignant
18.25	19.98	119.60	1040.0	0.09463	malignant
13.71	20.83	90.20	577.9	0.11890	malignant
13.00	21.82	87.50	519.8	0.12730	malignant
12.46	24.04	83.97	475.9	0.11860	malignant

Column	Feature Type	Sub-Type	Reasoning
mean radius	**Continuous**	Ratio	Physical measurement of the cell nucleus, can take any value within a range.
mean texture	**Continuous**	Ratio	Physical measurement (standard deviation of gray-scale values), can take any value within a range.
mean perimeter	**Continuous**	Ratio	Physical measurement, can take any value within a range.
mean area	**Continuous**	Ratio	Physical measurement, can take any value within a range.
mean smoothness	**Continuous**	Ratio	Local variation in radius lengths, a continuous measurement.
diagnosis	**Categorical**	Nominal (Binary)	Represents two distinct, unordered groups: malignant (cancerous) or benign (non-cancerous).

T20 Batting Performance Data

Player	Innings	Runs	Balls Faced	Strike Rate	Hundreds	Dismissal Type	Batting Hand
V Kohli	107	4008	2905	137.90	1	Caught	Right
R Sharma	140	3853	2772	138.60	4	Bowled	Right
B Azam	98	3698	2800	132.07	3	Caught	Right
M Rizwan	85	3135	2400	130.63	1	Run Out	Right
J Buttler	100	2720	1850	147.03	1	LBW	Right
D Warner	99	2894	2000	144.70	1	Caught	Left
C Gayle	79	1899	1220	155.65	2	Stumped	Left
K Williamson	87	2464	2050	120.19	0	Caught	Right
S Narine	92	1100	700	157.14	0	Caught	Left
F du Plessis	50	1675	1250	134.00	0	Bowled	Right

Column	Feature Type	Sub-Type	Reasoning
Player	**Categorical**	Nominal	A unique name/label used to group statistics.
Innings	**Discrete**	Ratio	A count of matches batted. Must be a whole number, and 0 is a true zero.
Runs	**Discrete**	Ratio	A count of runs scored. Must be a whole number, and 0 is a true zero.
Balls Faced	**Discrete**	Ratio	A count of balls faced. Must be a whole number, and 0 is a true zero.
Strike Rate	**Continuous**	Ratio	Calculated as $(Runs / Balls Faced) \times 100$. Can take any value within a range (includes decimals).
Hundreds	**Discrete**	Ratio	A count of innings scoring 100 or more. Must be a whole number.
Dismissal Type	**Categorical**	Nominal	Represents distinct, unordered groups (e.g., Caught, Bowled, LBW, Run Out).
Batting Hand	**Categorical**	Nominal (Binary)	Represents two distinct, unordered groups (Right, Left).

Feature Type	Sub-Type	Definition	Example (Dataset)	Formula (If Applicable)
1. Categorical	**Nominal** (or Binary)	Represents labels, names, or distinct, **unordered** groups. Numbers used are only for labeling (e.g., 0=no, 1=yes).	sex (male/female), diagnosis (malignant/benign), Dismissal Type	N/A (Based on grouping/labeling)
	Ordinal	Represents categories that have a **meaningful order or rank**. The interval between ranks is not necessarily equal or measurable.	pclass (1st, 2nd, 3rd), Review_Score (1 to 5), Quality (0-10 score)	N/A (Based on rank/order)
2. Discrete	**Ratio** (Count)	Data that are **countable** and must be **whole numbers**. The range of possible values is fixed (no values between integers). Zero is a true zero.	Innings, Runs, sibsp (siblings/spouses count)	N/A (Simple count of events)
3. Continuous	**Ratio**	Data that can take **any value** within a given range (decimals allowed). Zero means the complete absence of the property.	sepal_length, fare, alcohol, mean radius	Example: **Strike Rate** $= \frac{\text{Runs}}{\text{Balls Faced}} \times 100$
	Interval	Data that can take **any value** within a given range, but **zero is arbitrary** (it does not mean "nothing").	pH (A measure of acidity, not a count or true zero amount), Temperature (Celsius/Fahrenheit).	N/A (Based on an established scale without a true zero point)

Reference Links

- Python notebook
- How to Create Stunning Data Visualizations in Python: Top 10 Techniques to Learn
- Mastering Credit Risk Analysis: A Step-by-Step Guide to Descriptive Statistics in Python
- How Much Statistics Do You Really Need for Data Science?
- Phases of data science and analytics.

Data Distributions

Data Distributions Dataset

- ‘titanic’ dataset in seaborn

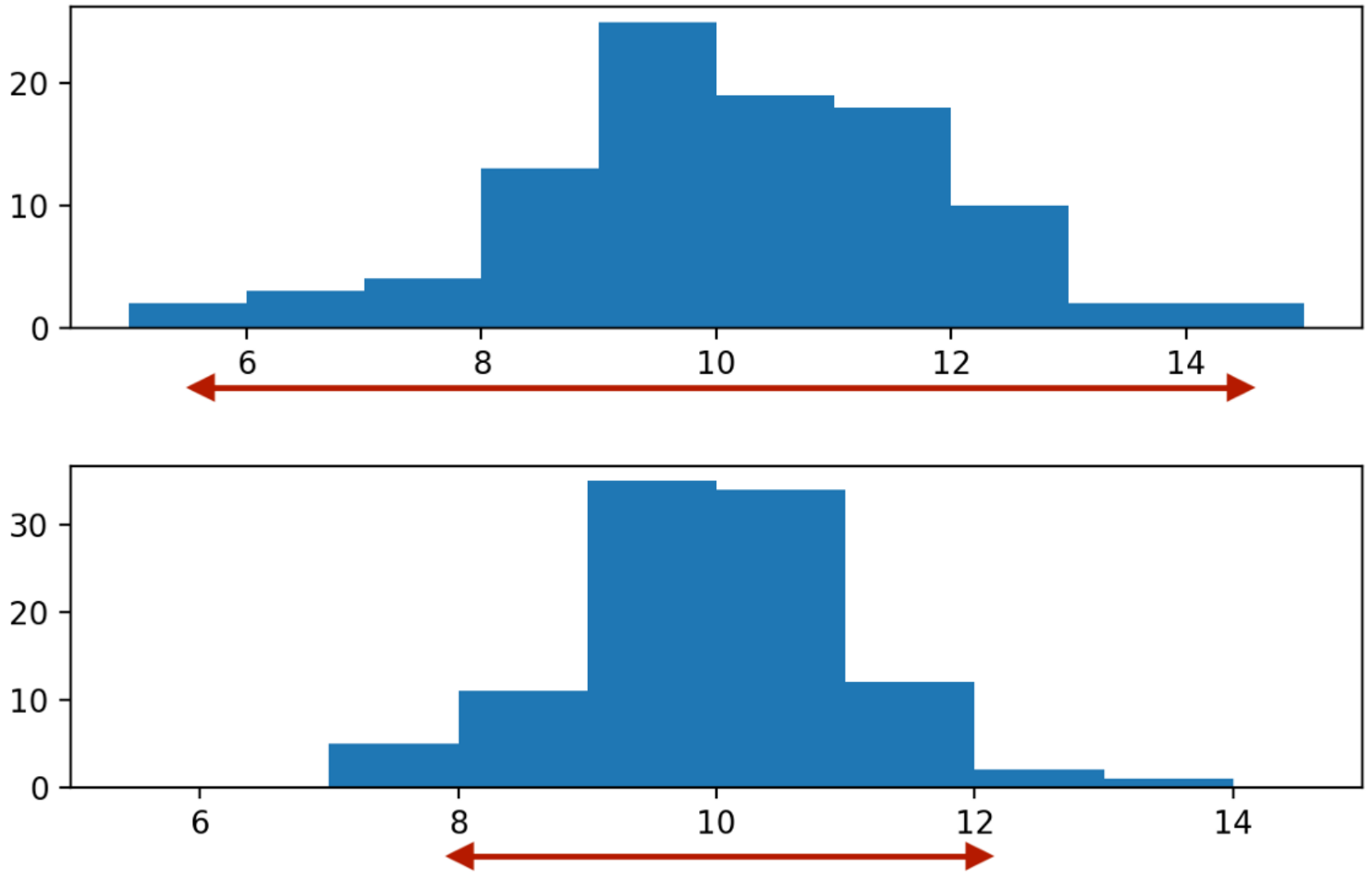
	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

Learning Objectives

By the end of this session, you will be able to:

- Understand the concept of data distributions
- Learn about skewness and kurtosis
- Apply Python code for calculating skewness and kurtosis
- Test your knowledge with quizzes and MCQs

What is Spread?



What is a Data Distribution?

A data distribution describes how values in a dataset are spread.

Types of distributions: Symmetric vs. Skewed

Examples: Normal, Uniform, Binomial, Poisson

Helps in: Statistical modeling, Machine learning, Data-driven decision-making

Types of Data Distributions

Distribution	Characteristics	Examples
Normal (Gaussian)	Bell-shaped, symmetric	Height, IQ scores
Uniform	All values equally likely	Dice rolls, Random numbers
Binomial	Success/failure trials	Coin flips, Pass/fail tests
Poisson	Rare events per time unit	Call center arrivals, Accidents
Exponential	Time until an event occurs	Service times, Waiting times

Real-World Applications of Distributions



FINANCE: STOCK RETURNS FOLLOW A NORMAL/LOG-NORMAL DISTRIBUTION



HEALTHCARE: DISEASE SPREAD FOLLOWS A POISSON DISTRIBUTION



RETAIL: CUSTOMER ARRIVALS FOLLOW AN EXPONENTIAL DISTRIBUTION



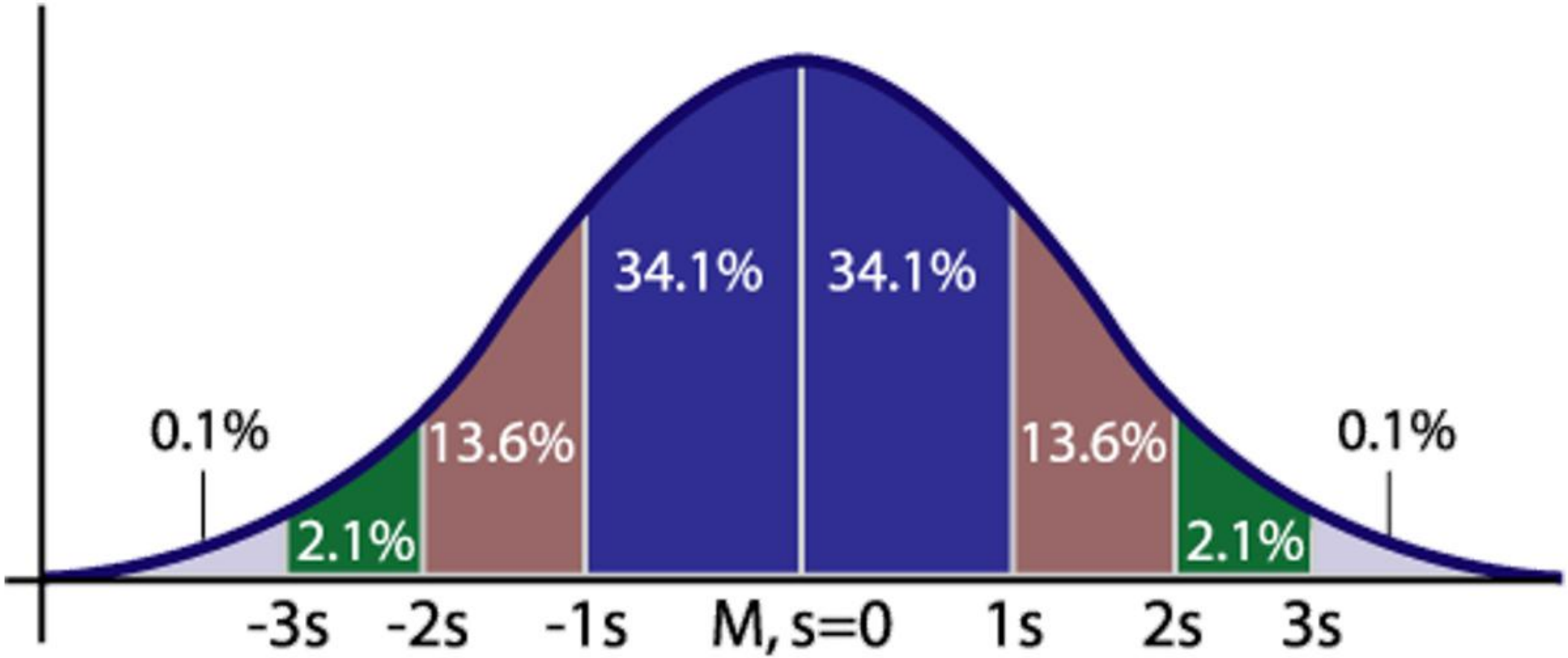
MANUFACTURING: QUALITY CONTROL USES A BINOMIAL DISTRIBUTION

Normal Distribution - The Bell Curve

- Symmetrical bell-shaped curve
- Defined by mean (μ) and standard deviation (σ)

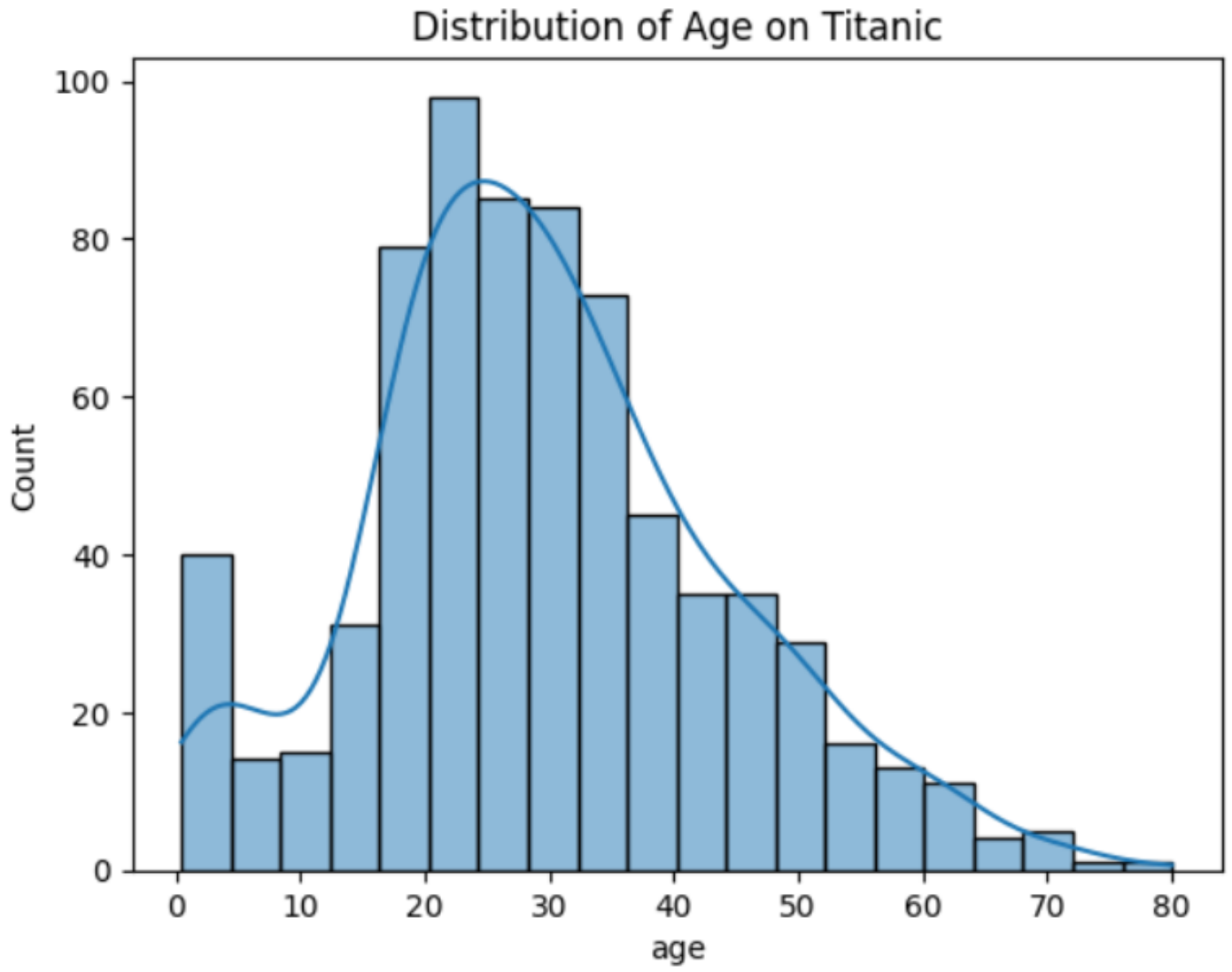
- 68-95-99.7 Rule:
    - 68% within 1σ
    - 95% within 2σ
    - 99.7% within 3σ

Normal Distribution - The Bell Curve



Visualizing Distribution

```
# Visualize the distribution of 'age'
sns.histplot(data['age'].dropna(), kde=True)
plt.title('Distribution of Age on Titanic')
plt.show()
```



Skewness - Symmetry of Data

Skewness tells us whether our data is skewed left, right, or symmetrical.

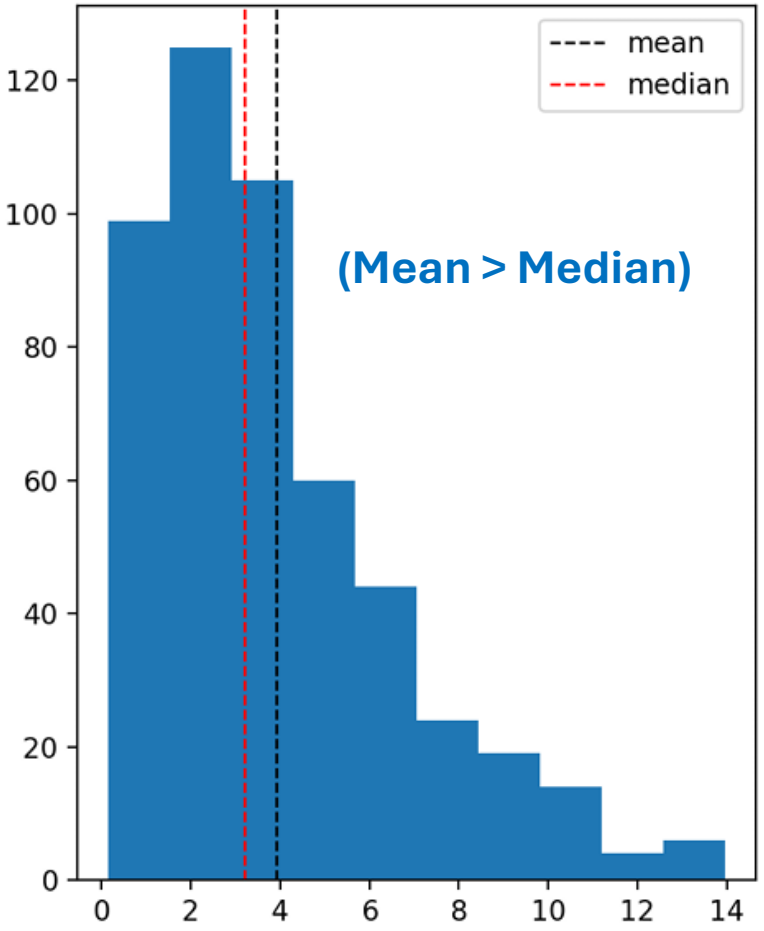
Measures asymmetry of distribution

- Formula:

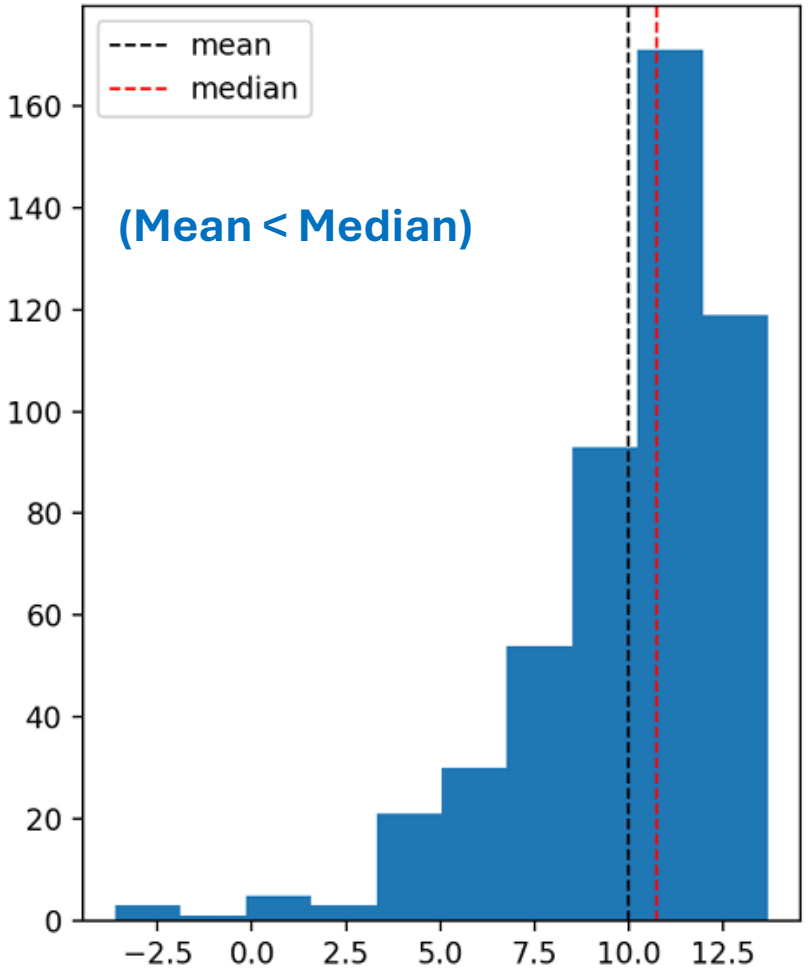
$$\text{Skewness} = \frac{n}{(n-1)(n-2)} \sum \left(\frac{x_i - \bar{x}}{s} \right)^3$$

Skewness - Symmetry of Data

Right-skewed



Left-skewed



Skewness Example

- Given a dataset: [150, 200, 250, 300, 350, 400, 800], compute the **skewness**.

$$\text{Skewness} = \frac{n}{(n-1)(n-2)} \sum \left(\frac{x_i - \bar{x}}{s} \right)^3$$

Skewness Example

- Given a dataset: [150, 200, 250, 300, 350, 400, 800], compute the **skewness**.

$$\text{Skewness} = \frac{n}{(n-1)(n-2)} \sum \left(\frac{x_i - \bar{x}}{s} \right)^3$$

Is the data right or left skewed?

Skewness Example

Dataset: [150, 200, 250, 300, 350, 400, 800]

- Mean:

$$\frac{150 + 200 + 250 + 300 + 350 + 400 + 800}{7} = 350$$

- **Median:** 300 (Middle value)
- **Skewness:** Since **Mean** > **Median**, data is **Right-Skewed**

Skewness in Python

```
# Calculate skewness
skewness_age = data['age'].skew()
skewness_fare = data['fare'].skew()

print(f"Skewness of Age: {skewness_age:.2f}")
print(f"Skewness of Fare: {skewness_fare:.2f}")
```

Explanation:

- **Positive skew (> 0):** Tail extends to the right.
- **Negative skew (< 0):** Tail extends to the left.
- **Zero skew:** Perfectly symmetrical.

Output:

```
Skewness of Age: 0.41
Skewness of Fare: 4.79
```

- **Age** is slightly positively skewed (some older passengers create a right tail).

- **Fare** is heavily positively skewed (some passengers paid extremely high fares).

Kurtosis - Tails of the Distribution

- Definition: **Kurtosis** measures whether data has **heavy or light tails**

- Types of kurtosis:

1. **Leptokurtic (High Kurtosis)**: Heavy tails (outliers). Kurtosis value > 3
2. **Mesokurtic (Normal Kurtosis)**: Like a normal distribution. Kurtosis value = 3 approx
3. **Platykurtic (Low Kurtosis)**: Light tails (less variation) Kurtosis value < 3

Formula:

$$\text{Kurtosis} = \frac{n(n+1)}{(n-1)(n-2)(n-3)} \sum \left(\frac{x_i - \bar{x}}{s} \right)^4 - \frac{3(n-1)^2}{(n-2)(n-3)}$$

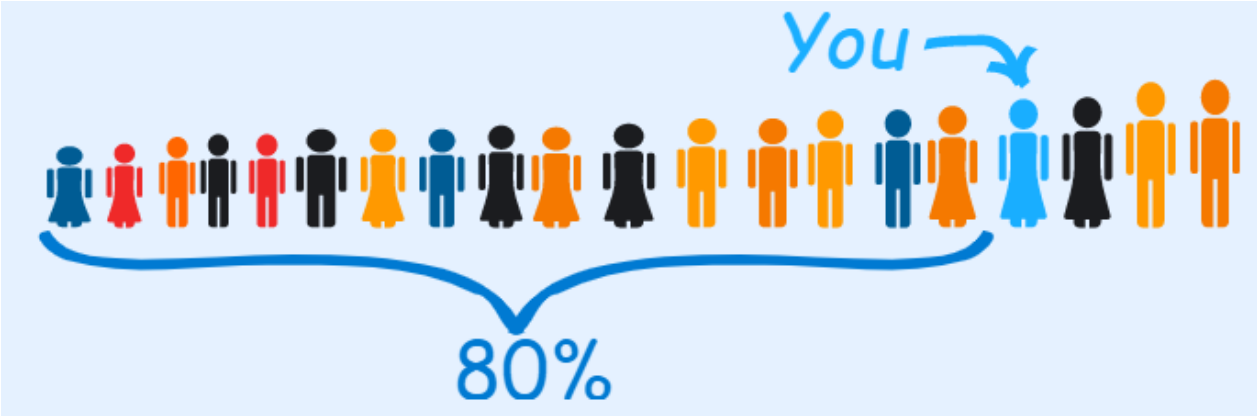
Skewness vs Kurtosis

- Skewness focusses on direction of outliers whereas kurtosis measures intensity of outliers.

- A dataset can be **symmetric (no skew)** but still have **high kurtosis** (many outliers)

Measures of Dispersion (IQR)

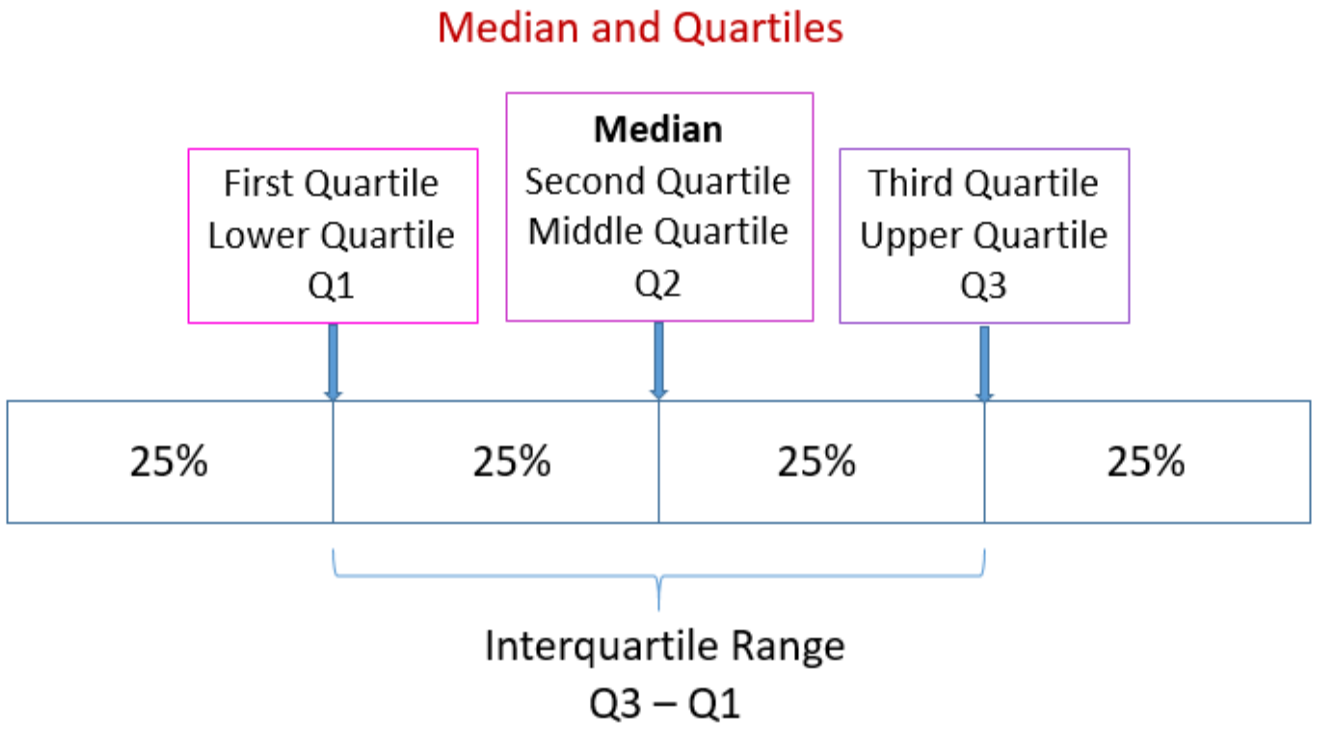
- **Percentile:** Percentiles indicate the percentage of scores that fall below a particular value.
- **Example:** Suppose you are the fourth tallest person in a group of 20 people. So that means 16 people are shorter than you. 80% of the people are shorter than you. That means you are at the 80th percentile.



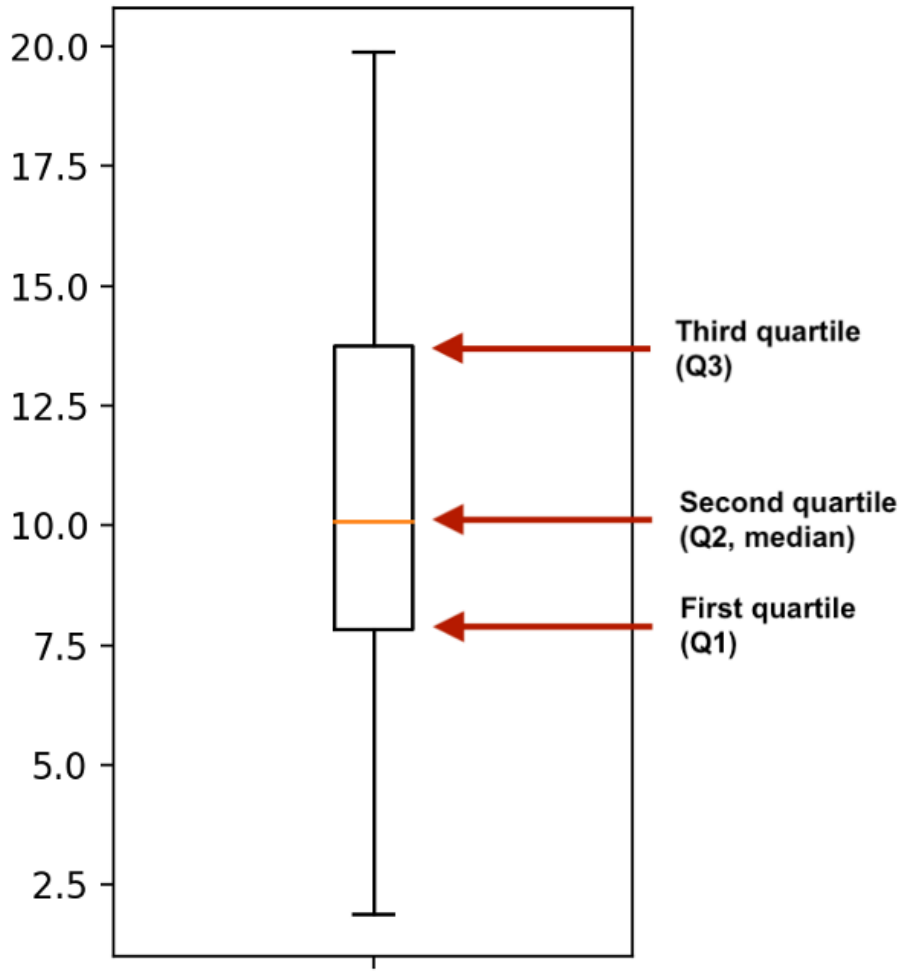
- Formula for Percentile = (Number of Values Below “x” / Total Number of Values) × 100

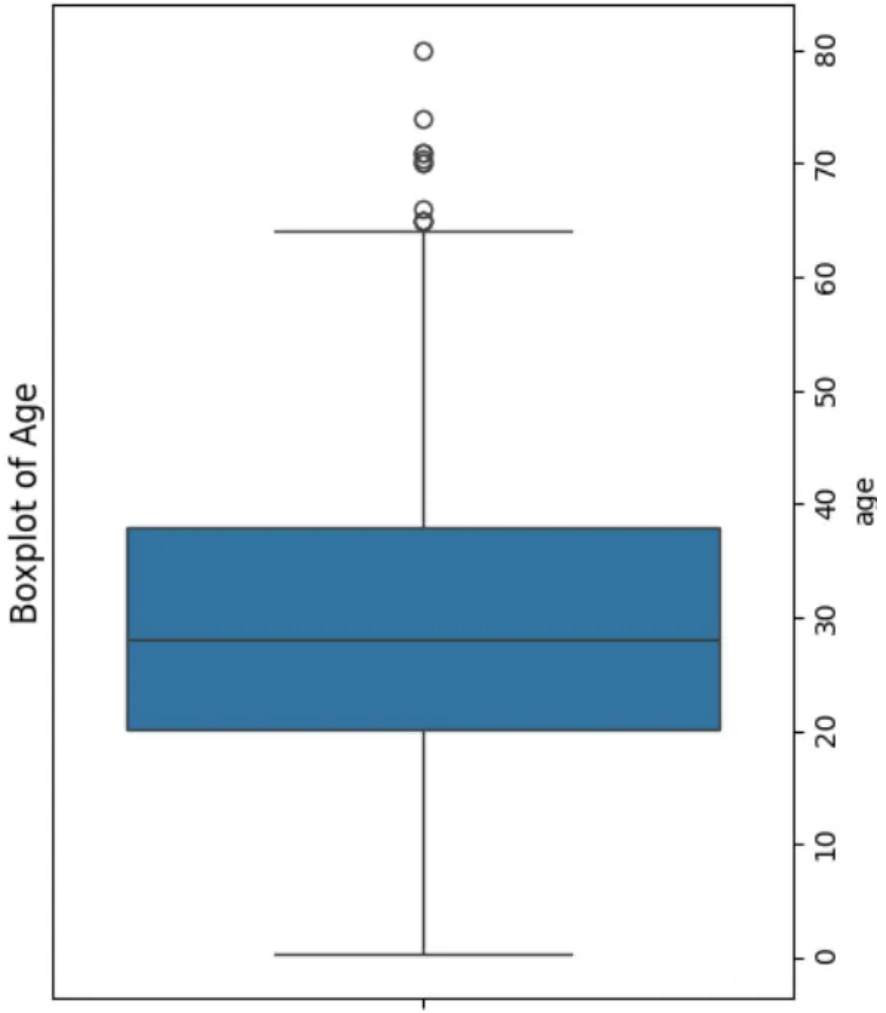
IQR (Interquartile range)

- **Quartiles:** Quartiles are the values that divide a list of numbers into quarters.
- Q1: Middle number between the smallest number and the median of the data set.
- Q2: Median of the data set.
- Q3: Middle value between the median and the highest value of the data set.
- Inter Quartile Range (IQR) = Q3 – Q1



Boxplot Representation





Python Implementation

https://github.com/vikashkmd/DSMLNotebooks/blob/main/Data%20distribution%20skewness%20and%20kurtosis.ipynb

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

[The Ultimate Guide to Data Distributions: Skewness, Centering, and Spread Made Simple]

Multiple Choice Questions (MCQs)

Q1:What does a right-skewed distribution indicate?
A) The majority of values are low, with a few high values
B) The majority of values are high, with a few low values
C) The data is perfectly symmetrical
D) The data has no variation

Q2:Which measure is best for skewed data?
A) Mean
B) Median
C) Mode
D) Standard deviation

Multiple Choice Questions (MCQs)

Q1:What does a right-skewed distribution indicate?
A) The majority of values are low, with a few high values ✅
B) The majority of values are high, with a few low values
C) The data is perfectly symmetrical
D) The data has no variation

Q2:Which measure is best for skewed data?
A) Mean
B) Median ✅
C) Mode
D) Standard deviation

Q3 . What does a high positive kurtosis (leptokurtic distribution) indicate about a dataset?

A) The data has lighter tails and fewer extreme outliers than a normal distribution.

B) The distribution is skewed heavily to the right.

C) The data has heavy tails and a higher risk of extreme outliers.

D) The mean, median, and mode are all equal.

Q3 . What does a high positive kurtosis (leptokurtic distribution) indicate about a dataset?

A) The data has lighter tails and fewer extreme outliers than a normal distribution.

B) The distribution is skewed heavily to the right.

C) The data has heavy tails and a higher risk of extreme outliers.

D) The mean, median, and mode are all equal.

Answer C)

In a negatively skewed (left-skewed) distribution, which of the following relationships generally holds true?

A) Mean > Median > Mode

B) Mean < Median < Mode

C) Mean = Median = Mode

D) Median > Mean > Mode

In a negatively skewed (left-skewed) distribution, which of the following relationships generally holds true?
A) Mean > Median > Mode
B) Mean < Median < Mode
C) Mean = Median = Mode
D) Median > Mean > Mode
Answer B)

Q. How does Winsorizing handle extreme outliers in a dataset?
A) It completely deletes the outlier observations from the dataset.
B) It replaces extreme values with the nearest non-outlier percentile values (e.g., 5th or 95th percentile).
C) It transforms the entire dataset using a logarithmic scale.
D) It replaces all missing and extreme values with the dataset mean.

Q. How does Winsorizing handle extreme outliers in a dataset?
A) It completely deletes the outlier observations from the dataset.
B) It replaces extreme values with the nearest non-outlier percentile values (e.g., 5th or 95th percentile).
C) It transforms the entire dataset using a logarithmic scale.
D) It replaces all missing and extreme values with the dataset mean.
Answer B)

Q. How is the Interquartile Range (IQR) calculated, and why is it preferred over standard deviation for skewed data?
A) IQR = Q3 - Q1; it is robust to extreme outliers.
B) IQR = Q4 - Q1; it covers 100% of the dataset.
C) IQR = Q3 + Q1; it measures the total variance of the sample.
D) IQR = Mean - Median; it measures the symmetry of the data.

Q. How is the Interquartile Range (IQR) calculated, and why is it preferred over standard deviation for skewed data?
A) IQR = Q3 - Q1; it is robust to extreme outliers.
B) IQR = Q4 - Q1; it covers 100% of the dataset.
C) IQR = Q3 + Q1; it measures the total variance of the sample.
D) IQR = Mean - Median; it measures the symmetry of the data.
Answer A)

Fill in the blanks

1. _____ measures the asymmetry of a data distribution.

2. The _____ rule states that 95% of data falls within 2 standard deviations of the mean in a normal distribution.

3. If the tails of a distribution are heavier than normal, it is called _____ kurtosis.

Fill in the blanks

1. _____ measures the asymmetry of a data distribution. (Answer: Skewness)

2. The _____ rule states that 95% of data falls within 2 standard deviations of the mean in a normal distribution. (Answer: 68-95-99.7 Rule)

3. If the tails of a distribution are heavier than normal, it is called _____ kurtosis. (Answer: Leptokurtic)

Session Summary

In this session, we have discussed about:

- Understand the concept of data distributions
- Learn about skewness and kurtosis
- Explore measures of central tendency
- Apply Python code for calculating skewness and kurtosis
- Test your knowledge with quizzes and MCQs

Reference Links

- Python notebook
- The Ultimate Guide to Data Distributions: Skewness, Centering, and Spread Made Simple
- How to Create Stunning Data Visualizations in Python: Top 10 Techniques to Learn
- Mastering Credit Risk Analysis: A Step-by-Step Guide to Descriptive Statistics in Python
- Phases of data science and analytics.

Probability Theory

Learning Objectives

By the end of this session, you will be able to:

- **Define** probability, sample space, and events.
- Understand key properties of probability.
- **Calculate** probabilities for dice, coins, and cards **with step-by-step solutions**.

What is Probability?

- Measures likelihood of an event (Range: 0 to 1).

$$\text{Probability of an event} = \frac{Number\ of\ favourable\ outcomes}{Total\ number\ of\ possible\ outcomes}$$

Basic Probability Concepts

- Probability – the chance that an uncertain event will occur (always between 0 and 1)
- Impossible Event – an event that has no chance of occurring (probability = 0)
- Certain Event – an event that is sure to occur (probability = 1)

Question: What is the probability of rolling a number greater than 4 on a fair six-sided die?



Problem: What is the probability of rolling a number greater than 4 on a fair six-sided die?

Solution: A six-sided die has six possible outcomes: 1, 2, 3, 4, 5, and 6. The favorable outcomes for rolling a number greater than 4 are 5 and 6.

- Number of favorable outcomes: 2 (which are 5 and 6)
- Total number of outcomes: 6

Answer: The probability of rolling a number greater than 4 is 2/6 = 1/3

Key Terms in Probability

Term	Meaning
Experiment	A procedure with uncertain outcome (e.g., rolling dice)
Sample Space (S)	Set of all possible outcomes
Event (E)	One or more outcomes from the sample space
Favorable Outcomes	Outcomes that make the event happen

Reference Dataset

- ‘titanic’ dataset crosstab

Gender	Did Not Survive (0)	Survived (1)	Total
Female	81	233	314
Male	468	109	577
Total	549	342	891

Simple Calculations

Find out the following probabilities:

1. Probability that a passenger is female (P(Female)).

2. Probability that a passenger is male (P(Male))

3. Probability that a passenger survived (P(Survived))

4. Probability that a passenger is female and survived (P(Female ∩ Survived))

Gender	Did Not Survive (0)	Survived (1)	Total
Female	81	233	314
Male	468	109	577
Total	549	342	891

Probability that a passenger is female (P(Female))

$$P(Female) = \frac{\text{Total number of females}}{\text{Total passengers}} = \frac{314}{891} \approx 0.3524$$

Probability that a passenger is male (P(Male))

$$P(Male) = \frac{577}{891} \approx 0.6476$$

Probability that a passenger survived (P(Survived))

$$P(Survived) = \frac{342}{891} \approx 0.3838$$

$$P(Female \cap Survived) = \frac{233}{891} \approx 0.2614$$

Properties of Probability

1. Non-negativity

Every probability is a non-negative number, ranging from 0 to 1. This means you cannot have a probability less than 0 or greater than 1.

$$0 \leq P(A) \leq 1$$

for any event A.

2. Normalization

The probability of the sample space S, which represents all possible outcomes of a random experiment, is 1.

$$P(S) = 1$$

Complement Rule

The probability of the complement of an event A (denoted as A^c or $\bar{A}$)—meaning the event that A does not occur—is 1 minus the probability that A does occur.

$$P(A^c) = 1 - P(A)$$

Problem: If the probability of raining tomorrow is 0.3, what is the probability of it not raining?

Probability that a passenger survived (P(Survived))

$$P(Survived) = \frac{342}{891} \approx 0.3838$$

Addition Rule

If two events A and B are not mutually exclusive (meaning they can occur together), the probability that A or B occurs is the sum of their individual probabilities minus the probability of their intersection (the probability that both events occur).

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Problem: What is the probability of drawing a red card or a king from a standard deck of cards?

Problem: What is the probability of drawing a red card or a king from a standard deck of cards?

Solution: $\frac{1}{2} + \frac{1}{13} - \frac{1}{26} = \frac{7}{13}$.

Explanation: Red cards and kings are not mutually exclusive (there are 2 red kings), so subtract the intersection.

Probability that a passenger is female or survived (P(Female ∪ Survived))

Using the **Addition Rule**:

$$P(Female \cup Survived) = P(Female) + P(Survived) - P(Female \cap Survived)$$

Probability that a passenger is female or survived (P(Female ∪ Survived))

$$P(Female \cup Survived) = P(Female) + P(Survived) - P(Female \cap Survived)$$

$$= \frac{314}{891} + \frac{342}{891} - \frac{233}{891} = \frac{423}{891} \approx 0.4748$$

Addition Rule for Mutually Exclusive Events

If two events A and B are mutually exclusive (meaning they cannot both occur at the same time), the probability that A or B occurs is the sum of their individual probabilities.

$$P(A \cup B) = P(A) + P(B)$$

This can be extended to any number of mutually exclusive events.

Problem: What is the probability of rolling a 1 or 2 on a six-sided die?

Problem: What is the probability of rolling a 1 or 2 on a six-sided die?

Solution: $\frac{1}{6} + \frac{1}{6} = \frac{1}{3}$.

Explanation: Rolling a 1 and rolling a 2 are mutually exclusive, so you add their probabilities.

Conditional Probability

The probability of an event A given that event B has occurred (denoted $P(A|B)$) is the probability of the intersection of A and B divided by the probability of B, provided that $P(B)$ is not zero.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Probability that a passenger survived given that they are female (P(Survived | Female))

Conditional Probability

Probability that a passenger survived given that they are female (P(Survived | Female))

This uses **Conditional Probability**:

$$P(Survived|Female) = \frac{P(Female \cap Survived)}{P(Female)}$$

Gender	Did Not Survive (0)	Survived (1)	Total
Female	81	233	314
Male	468	109	577
Total	549	342	891

Conditional Probability

This uses **Conditional Probability**:

$$P(Survived|Female) = \frac{P(Female \cap Survived)}{P(Female)}$$

Gender	Did Not Survive (0)	Survived (1)	Total
Female	81	233	314
Male	468	109	577
Total	549	342	891

$$P(Survived|Female) = \frac{P(Female \cap Survived)}{P(Female)} = \frac{233}{314} \approx 0.7417$$

Conditional Probability

Probability that a passenger survived given that they are male (P(Male | Female))

Gender	Did Not Survive (0)	Survived (1)	Total
Female	81	233	314
Male	468	109	577
Total	549	342	891

Conditional Probability

Probability that a passenger survived given that they are male (P(Male | Female))

$$P(Survived|Male) = \frac{109}{577} \approx 0.1889$$

Interpretation: Given the passenger is male, there is only an **18.89% chance** he survived.

Baye's Theorem

Bayes' Theorem is a way to update our beliefs about the probability of an event based on new evidence.

In simple terms:
It tells us how likely something is, *given* that we already know something else has happened.

Baye's theorem

Formula of Bayes' Theorem

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Where:

- $P(A|B)$: **Posterior probability** (Probability of A *given* B has occurred)
- $P(B|A)$: **Likelihood** (Probability of B *given* A is true)
- $P(A)$: **Prior probability** (Initial belief about A)
- $P(B)$: **Marginal probability** (Total probability of B happening)

Baye's theorem example

What's the probability a survivor was female?

That's $P(Female \mid Survived)$

We use **Bayes' Theorem**:

$$P(Female|Survived) = \frac{P(Survived|Female) \cdot P(Female)}{P(Survived)}$$

$$= \frac{0.742 \cdot 0.352}{0.384} \approx \frac{0.261}{0.384} \approx 0.682$$

> So around **68.2% of the survivors were women**. Bayes for the win!

Difference

Feature	Conditional Probability	Bayes' Theorem
Definition	Probability of A given B	Updates probability of A after B is observed
Known Values	(P(A and B)), (P(B))	(P(B))
Purpose	Direct computation	Reversing or updating beliefs
Focus	Descriptive	Inference / learning from data
Use Case	"If it's raining, what's the chance the ground is wet?"	"The ground is wet — what's the chance it rained?"

Multiple Choice Questions (MCQs)

Q1: If events A and B are independent, $P(A \cap B)$ is:
A) $P(A) + P(B)$
B) $P(A) – P(B)$
C) $P(A) \times P(B)$
D) None of the above

Q2: What is the sample space for a coin toss?
A) {H, T}
B) {1, 2}
C) {Red, Black}

Multiple Choice Questions (MCQs)

Q1: If events A and B are independent, P(A ∩ B) is:
A) P(A) + P(B)
B) P(A) – P(B)
C) P(A) × P(B) ✅
D) None of the above

Q2: What is the sample space for a coin toss?
A) {H, T}
B) {1, 2}
C) {Red, Black}
Answer: A) {H, T}

Session Summary

In this session, we have discussed about:

- **Define** probability, sample space, and events.
- Understand key properties of probability.
- **Calculate** probabilities for dice, coins, and cards **with step-by-step solutions**.

Reference Links

- 10 Must-Know Probability Concepts for Every Aspiring Statistician and Data Scientist
- Mastering Credit Risk Analysis: A Step-by-Step Guide to Descriptive Statistics in Python
- 10 Statistical Concepts Every Aspiring Data Scientist Should Know — The Davangers' Guide!
- Top 20 Interview Questions on Descriptive and Inferential Statistics
- 10 Must-Know Statistics Interview Questions (With Real Examples & Formulas!)

Common Probability Distributions

Learning Objectives

By the end of this session, you will be able to:

- Understand the concept of random variable and probability distributions.
- Understand Normal, Binomial, and Poisson Distributions.
- Learn the formulas, attempt quizzes and solve numerical problems.
- How to work with probability distributions using Python!

Random Variable Example

- Number of sixes that will be hit in the last over of a T20 match?

Random Variable Example

- Number of sixes that will be hit in the last over of a T20 match?

No of sixes	0	1	2	3	4	5	6

- A random variable is a variable that takes on different values based on the outcome of a random event.

- It can be thought of as a function that assigns numerical values to each possible outcome of a random process.

Types of Random Variable

- **Discrete Random Variable:** Takes on a countable number of distinct values (e.g., number of heads in a series of coin flips).

- **Continuous Random Variable:** Takes on an infinite number of possible values within a given range (e.g., the exact height of students in a class).

Types of Distributions

Discrete Distributions: Binomial, Poisson

Continuous Distributions: Normal, Exponential

What is a Probability Distribution?

- A probability distribution is a function that describes the likelihood of outcomes in a sample space.

- Examples include: Normal, Binomial, Poisson

What is a Probability Distribution?

Ex: Consider the random experiment of tossing two coins.

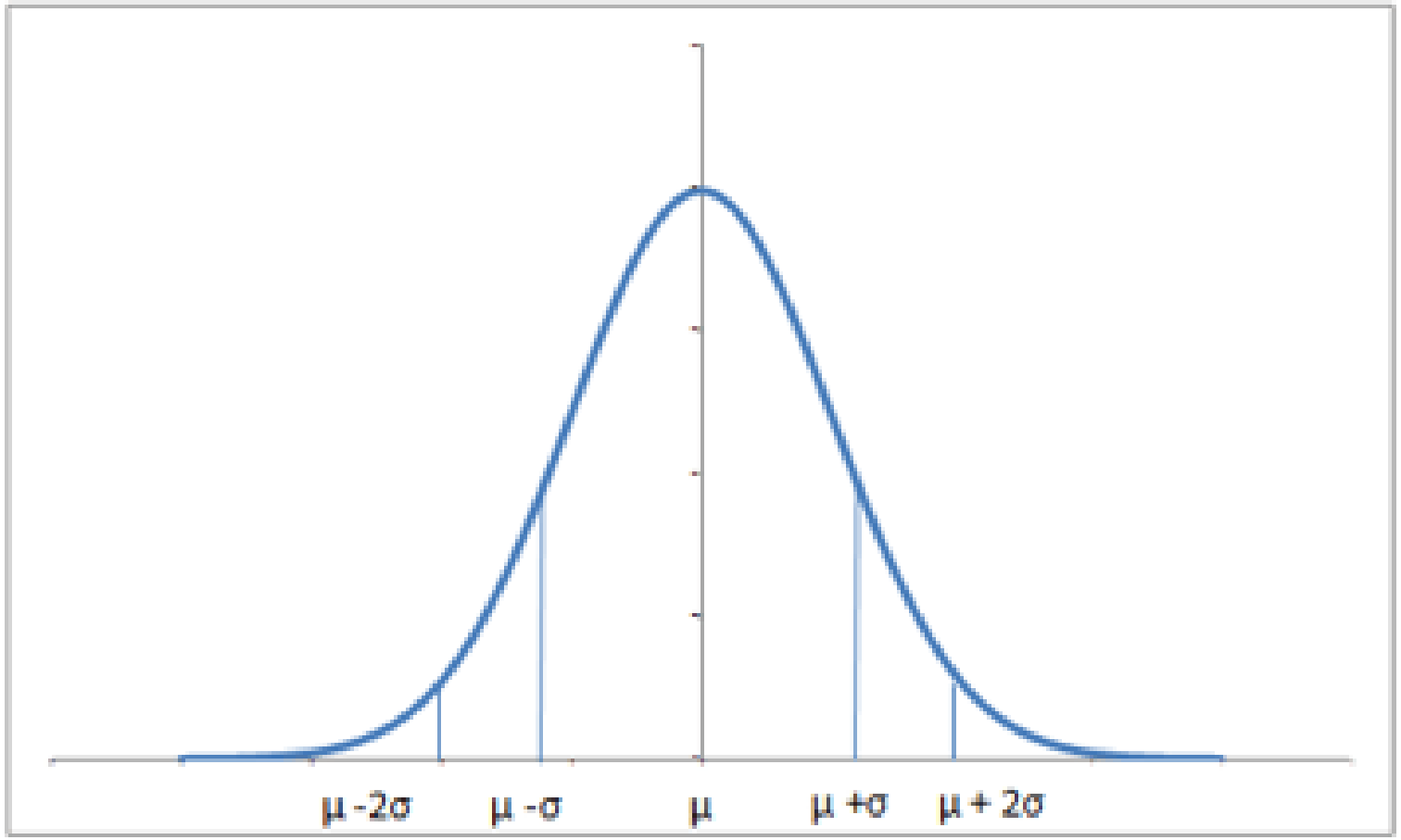
Let the occurrence of heads to be counted. There may be 0, 1 or 2 heads.

Possible outcomes	TT	HT	TH	HH
Number of heads	0	1	1	2
Probability	¼	¼	¼	¼

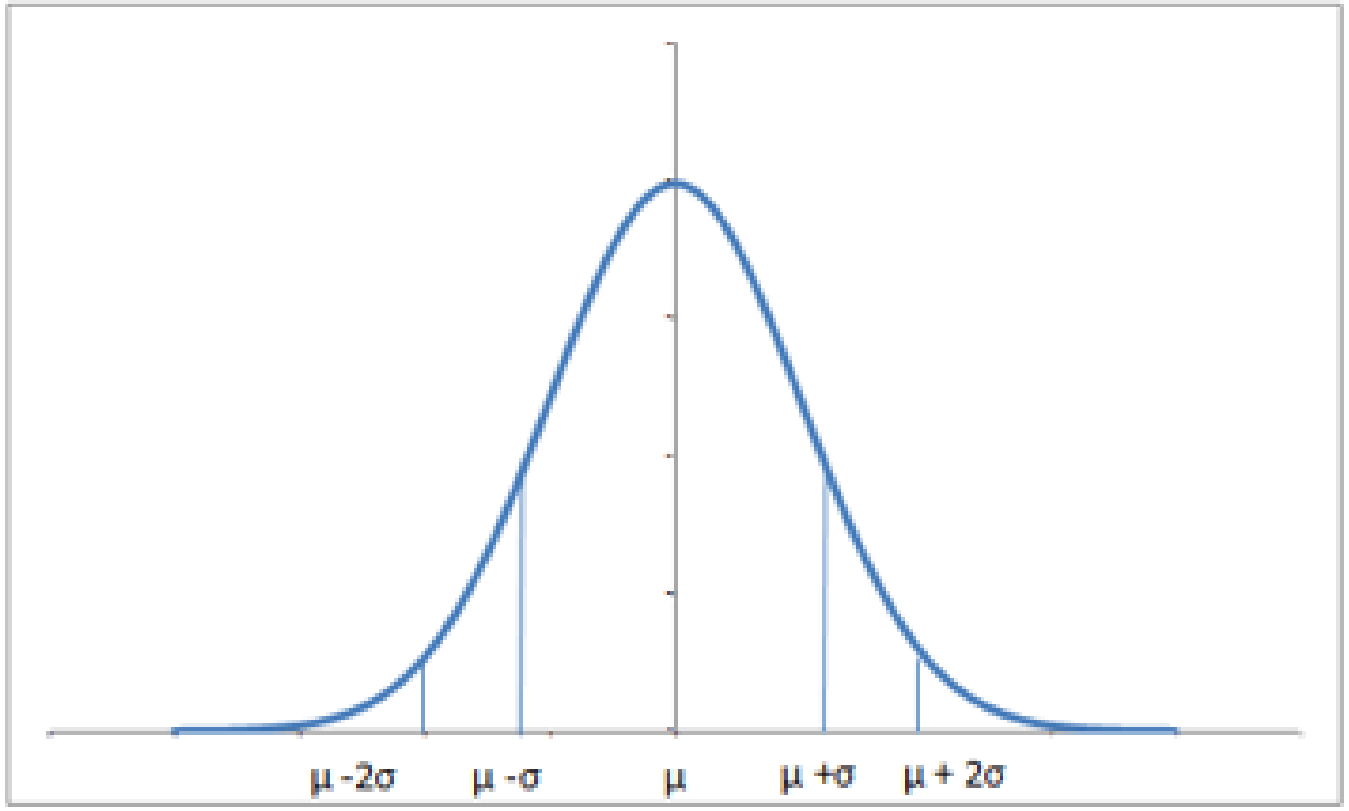
A probability distribution describes how the values of a random variable are distributed. It provides a complete description of the likelihood of different outcomes.

Normal Distribution

- A continuous distribution with a symmetric, bell-shaped curve.
- Defined by mean (μ) and standard deviation (σ).



Normal Distribution - formula



$$f(x|\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- Mean = Median = Mode
- 68-95-99.7 Rule
- Symmetrical shape

Normal Distribution - Example

- If height ~ N(170, 10), what is the probability a person is between 160 and 180 cm?

Normal Distribution – Example Solution

- If height ~ N(170, 10), what is the probability a person is between 160 and 180 cm?

- $Z = (x - \mu)/\sigma \rightarrow Z = (180 - 170)/10 = 1$
- $P(-1 < Z < 1) = 68\%$
- Hence, 68% of people are between 160 and 180 cm

Binomial Distribution

- Models the number of successes in n independent trials with probability p.
- Used for yes/no outcomes.

The probability mass function (PMF) for a Binomial distribution is:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Where:

- $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ is the binomial coefficient, representing the number of ways to choose k successes out of n trials.
- p^k is the probability of having k successes.
- $(1-p)^{n-k}$ is the probability of having $n-k$ failures.

Binomial Distribution Properties

- Discrete distribution
- Fixed number of trials (n)
- Constant probability (p)
- Independent trials

Binomial Distribution example

In an inspection of cement companies 60% of all industries had emissions that do not meet the pollution requirement. For a random sample of 15 companies, compute the following

1. Probability that all 15 fails the inspection
2. Exactly 8 fails the inspection

Binomial Distribution example

In an inspection of cement companies 60% of all industries had emissions that do not meet the pollution requirement. For a random sample of 15 companies, compute the following

1. Probability that all 15 fails the inspection
2. Exactly 8 fails the inspection

Given:

n = 15

p = 0.60

q = 1-p = 1-0.6 = 0.4

Probability that all 15 fails the inspection

Here x = 15

$P(x) = {}^{n}C_{x}\, p^{x} q^{n-x}$, x: 0,1,2….,n

$P(x=15) = {}^{15}C_{15}\, (0.6)^{15} (0.4)^{0}$

P(X=15) = 0.00047

Binomial Distribution example

In an inspection of cement companies 60% of all industries had emissions that do not meet the pollution requirement. For a random sample of 15 companies, compute the following

1. Probability that all 15 fails the inspection
2. Exactly 8 fails the inspection

Given:

n = 15

p = 0.60

q = 1-p = 1-0.6 = 0.4

Probability that exactly 8 fails the inspection

Here x = 8

$P(x) = {}^{n}C_{x}\, p^{x} q^{n-x}$, x: 0,1,2….,n

$P(x=8) = {}^{15}C_{8}\,(0.6)^{8}(0.4)^{15-8}$

$P(x=8) = {}^{15}C_{8}\,(0.6)^{8}(0.4)^{7}$

P(X=8) = 0.17708

Poisson Distribution

Sometimes there may be a situation that out of a large number of trials, probability of getting a success may be very small i.e., the event may be a rare event.

Ex: Cyber attack, Fraud, Probability of a tree falling on someone's head.
In this kind of situation where the occurrence of event is very rare Poisson distribution may be used.

Poisson distribution is the limiting case of binomial distribution. Under the following conditions binomial distribution tend to Poisson distribution.

1. There are infinite number of trials; $n \rightarrow \alpha$
2. Probability of getting an event is very small $p \rightarrow 0$
3. The product 'np' is constant $np = \lambda$

Poisson Distribution

If X is a random variable following a Poisson distribution, it is denoted as $X \sim \text{Poisson}(\lambda)$, where λ is the average number of occurrences in the interval.

The probability mass function (PMF) for a Poisson distribution is:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

Where:

- λ is the average rate of occurrence.
- e is the base of the natural logarithm (approximately 2.71828).
- k is the number of occurrences.

Properties of Poisson Distribution

1. Mean $= \lambda$
2. Variance $= \lambda$
3. In Poisson distribution Mean $=$ Variance
4. Standard deviation $=$ Sqrt(Variance)

Example

Suppose a bookstore receives an average of 3 customer inquiries per hour. The number of inquiries in a given hour follows a Poisson distribution with $\lambda = 3$.

Let's calculate the probability of receiving exactly 4 inquiries in an hour:

$$P(X = 4) = \frac{3^4 e^{-3}}{4!} = \frac{81 \times 0.0498}{24} = 0.168$$

So, the probability of receiving exactly 4 inquiries in an hour is approximately 0.168.

How to perform Probability distributions using Python?

Comparison



BINOMIAL: FIXED TRIALS, YES/NO



POISSON: RARE EVENTS OVER TIME



NORMAL: CONTINUOUS DATA, BELL CURVE

Multiple Choice Questions (MCQs)

What is a Random Variable?

a) A variable that takes deterministic values.
b) A variable that always takes integer values.
c) A constant that never changes.
d) A variable that assigns numerical values to the outcomes of a random process.

Multiple Choice Questions (MCQs)

What is a Random Variable?

a) A variable that takes deterministic values.
b) A variable that always takes integer values.
c) A constant that never changes.
d) A variable that assigns numerical values to the outcomes of a random process.

Multiple Choice Questions (MCQs)

Q1: Which distribution has mean = variance?

A) Normal

B) Binomial

C) Poisson

Q2: Which is used for yes/no trials?

A) Poisson

B) Binomial

Multiple Choice Questions (MCQs)

Q1: Which distribution has mean = variance?

A) Normal

B) Binomial

C) Poisson ✅

Q2: Which is used for yes/no trials?

A) Poisson

B) Binomial ✅

Session Summary

In this session, we have discussed about:

- Understand the concept of random variable and probability distributions.
- Understand Normal, Binomial, and Poisson Distributions.
- Learn the formulas, attempt quizzes and solve numerical problems.
- How to work with probability distributions using Python!

Reference Links

- Top 15 Probability Distribution Questions for Data Science Interviews
- Analyzing loan data with binomial and poisson distribution
- Notebook
- 10 Must-Know Probability Concepts for Every Aspiring Statistician and Data Scientist
- Top 20 Interview Questions on Descriptive and Inferential Statistics
- 10 Must-Know Statistics Interview Questions (With Real Examples & Formulas!)

Types of sampling methods

Learning Objectives

By the end of this session, you will be able to:

- Understand the concept of sampling.
- Understand different types of sampling and their use case.
- Implementation of sampling techniques in Python!

What is Sampling?

- Selecting a subset from a larger population.
- Saves time, cost
- Enables conclusions.

Why is Sampling Important?

- ✅ Populations too large to fully study
- ✅ Reduces effort
- ✅ Provides valid results

Random Sampling

- Each member has equal chance.

- Simple Random Sampling is used when you want a straightforward and fair representation of the entire population.

$$P(\text{select an item}) = \frac{n}{N}$$

- Randomly picking 10 passengers from 100 Titanic passengers. Each has a 10% chance of being selected.

Stratified Sampling

- In **stratified sampling**, the population is
  - ✓ **divided into groups (strata)**
  - ✓ based on an important characteristic (like gender), and
  - ✓ then **samples are drawn proportionally from each group**.
- This ensures your sample **respects the structure** of the population.
- Formula: (stratum size / population size) × sample size.

Stratified Sampling

- A college has 800 students: 600 UG, 200 PG.

You want to sample 100 students using stratified sampling technique. How many students from UG and PG will be in your sample?

Stratified Sampling

- A college has 800 students: 600 UG, 200 PG.

You want to sample 100 students using stratified sampling technique. How many students from UG and PG will be in your sample?

Undergrad sample =

$$\frac{600}{800} \times 100 = 75$$

Postgrad sample =

$$\frac{200}{800} \times 100 = 25$$

Systematic Sampling

- In **systematic sampling**, you **select every k-th item** from an ordered list.
- It’s simple, fast, and effective when your data is organized.
- Select every k-th element.
- Formula: $k = N / n$
- Example: Survey every 10th customer entering a store.

Cluster Sampling

- Divide into clusters.
- Randomly select whole clusters.
- Example: Randomly selecting 2 out of 5 schools and surveying **all students** inside.

Comparison

Sampling Method	Key Idea	Best Use Case
Random Sampling	Equal chance for all	General surveys, experiments
Stratified Sampling	Group by characteristic, then sample	When important subgroups exist
Systematic Sampling	Every k-th item selected	Assembly lines, orderly populations
Cluster Sampling	Randomly pick entire groups	When groups are naturally formed

Python Implementation

MCQ

1. Which sampling ensures proportional representation?

A) Random

B) Stratified

C) Cluster

2. Which method selects every k-th item?

A) Random

B) Systematic

C) Cluster

MCQ

1. Which sampling ensures proportional representation?

A) Random

B) Stratified ✅

C) Cluster

2. Which method selects every k-th item?

A) Random

B) Systematic ✅

C) Cluster

Session Summary

In this session, we have discussed about:

- Understand the concept of sampling.
- Understand different types of sampling and their use case.
- Implementation of sampling techniques in Python!

Reference Links

- Complete Guide to Sampling Methods: Random, Stratified, Systematic, and Cluster Sampling with Python Examples
- Analyzing loan data with binomial and poisson distribution
- 10 Must-Know Probability Concepts for Every Aspiring Statistician and Data Scientist
- Top 20 Interview Questions on Descriptive and Inferential Statistics
- 10 Must-Know Statistics Interview Questions (With Real Examples & Formulas!)

Hypothesis Formulation

Learning Objectives

By the end of this session, you will be able to:

- Basics of Hypothesis Testing
- Steps of Hypothesis Testing
- Formulate H_0 and H_1
- Types of Error
- Significance Level
- Types of Statistical Tests

What is a Hypothesis and Hypothesis Testing?

- An assumption or educated guess about a population parameter.
- Tested using sample data.

- Hypothesis testing is like a detective game where you :

1. start with an assumption (a hypothesis);
2. then collect evidence (data) to decide whether
3. your assumption is likely to be true or not.

- It’s a way of making decisions or inferences about a population based on a sample of data.

Steps in Hypothesis Testing

1

1. Formulate H_0 and H_1

2

2. Choose α

3

3. Select Test

4

4. Calculate Test Statistic

5

5. Make Decision

Types of Hypotheses

✅ Null Hypothesis (H_0): This is the hypothesis that there is no effect or no difference. It's the status quo or the assumption we want to test against.

✅ Alternative Hypothesis (H_1): Opposite of null hypothesis.

When the null hypothesis is rejected the alternate hypothesis is accepted and vice versa.

Null and Alternate Hypothesis

- Null Hypothesis:
  - Hypothesis of no difference (Status Quo)
- Alternate hypothesis:
  - Mutually exclusive statement of Null hypothesis

H_0	H_a
equal (=)	not equal (≠) **or** greater than (>) **or** less than (<)
greater than or equal to (≥)	less than (<)
less than or equal to (≤)	more than (>)

Examples

Problem: Is the new drug better?

✅ H_0: New drug = Old drug

✅ H_1: New drug ≠ Old drug

Company claims average salary = ₹50,000/month.

✅ H_0: $\mu = 50,000$

✅ H_1: $\mu \neq 50,000$

Examples

Construct the null and alternative hypothesis for the following situation:

Your mobile bill per month is INR 600 per month with a standard deviation of INR 50.

This month you received a bill of INR 750.

You feel that the bill was high and you want to investigate if there is any reason for concern.

Examples

Construct the null and alternative hypothesis for the following situation:

Your mobile bill per month is INR 600 per month with a standard deviation of INR 50.

This month you received a bill of INR 750.

You feel that the bill was high and you want to investigate if there is any reason for concern.

$$H_0: \mu \leq INR600$$

$$H_1: \mu > INR600$$

Steps in Hypothesis Testing

1

1. Formulate H_0 and H_1

2

2. Choose α

3

3. Select Test

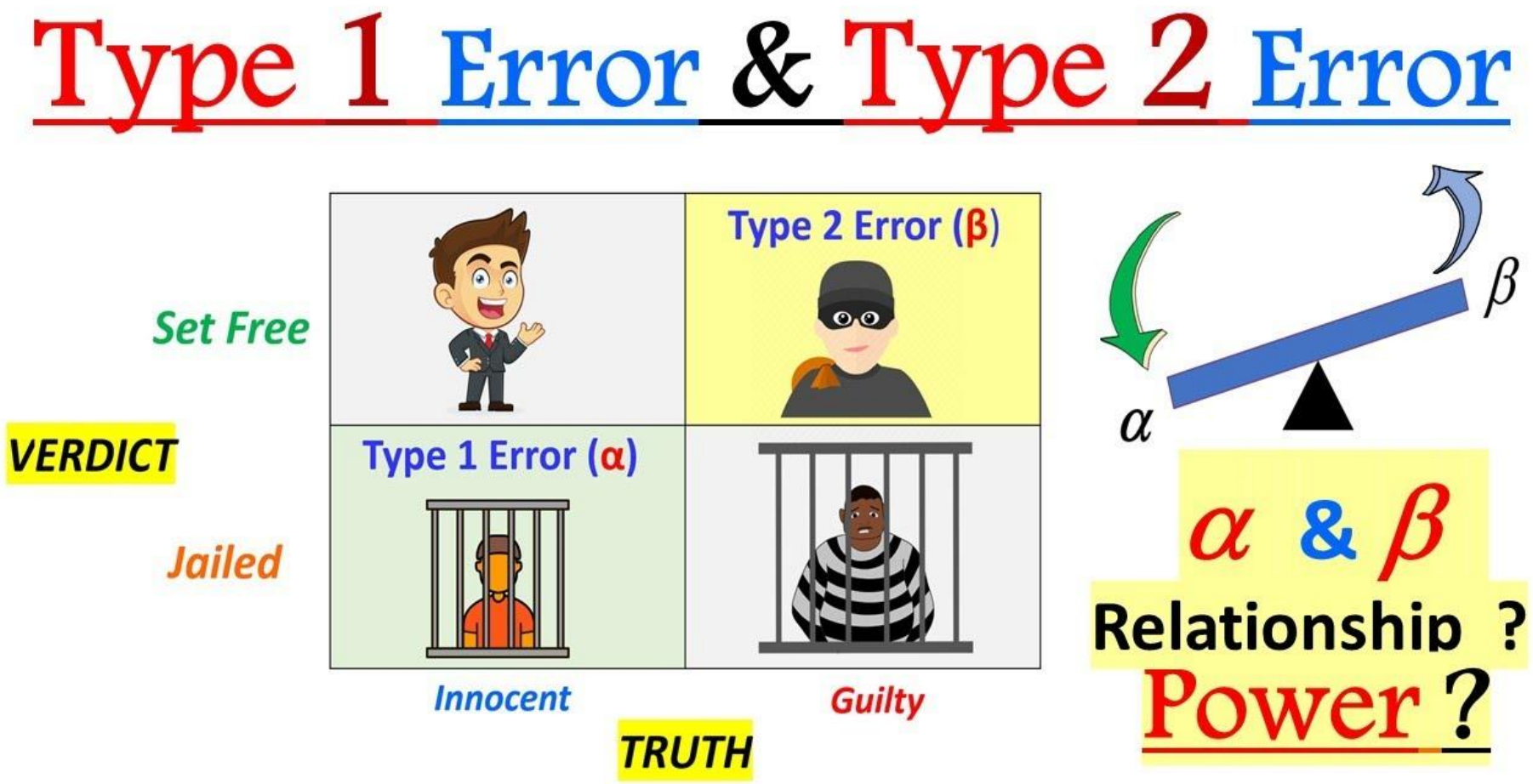
4

4. Calculate Test Statistic

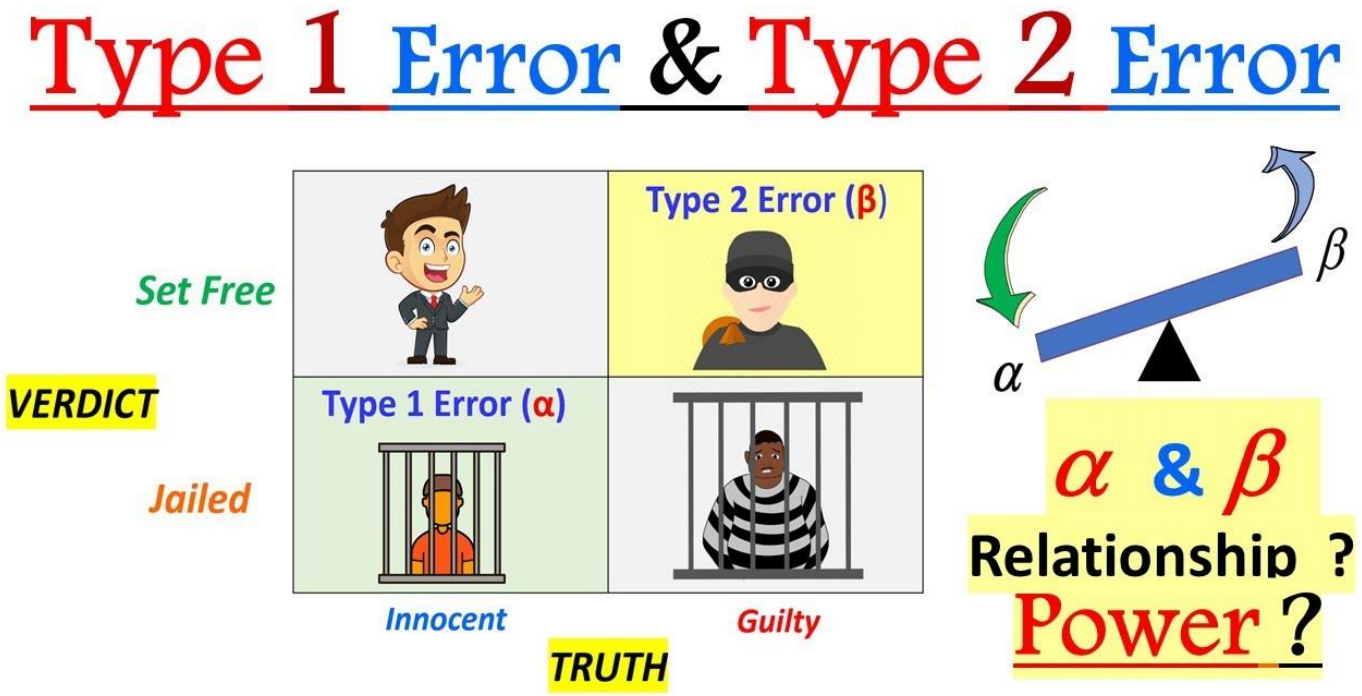
5

5. Make Decision

Significance level – Type 1 and Type 2 Error



Significance level – Type 1 and Type 2 Error



Level of significance (alpha) is the probability of rejecting the null hypothesis when it is true.

$\alpha = 0.01$ or 1%

$\alpha = 0.05$ or 5%

$\alpha = 0.10$ or 10%

Steps in Hypothesis Testing

1

1. Formulate H_0 and H_1

2

2. Choose α

3

3. Select Test

4

4. Calculate Test Statistic

5

5. Make Decision

Types of Tests

A **one-tail test** tests for the possibility of the relationship in one direction, either greater than or less than a certain value. **Directional (> or <)**

A **two-tail test** tests for the possibility of the relationship in both directions, whether it's significantly greater or less than a certain value. **Non-directional (≠)**

Two tail	Left tail	Right tail
H0: Mean=10 Ha: Mean≠10	H0: Mean ≥ 10 Ha: Mean < 10	H0: Mean ≤ 10 Ha: Mean >10

Steps in Hypothesis Testing

1

1. Formulate H_0 and H_1

2

2. Choose α

3

3. Select Test

4

4. Calculate Test Statistic

5

5. Make Decision

Test Statistic

A **test statistic** is a standardized value derived from sample data during a hypothesis test.

It helps determine whether to reject the null hypothesis.

Different tests (t-test, chi-square, etc.) have different formulas for calculating their respective test statistics.

$$\textbf{Test statistics} = \frac{\textit{Sample statistic} - \textit{Population Parameter}}{\textit{Standard Error}}$$

Test Statistic

We want to test that the population mean is different than 10

We want to test that the population mean is less than 10

We want to test that the population mean is greater than 10

Two tail	Left tail	Right tail
H0: Mean=10 Ha: Mean≠10	H0: Mean ≥ 10 Ha: Mean < 10	H0: Mean ≤ 10 Ha: Mean >10

Commonly used Critical z values

Significance Level (α)	Two-Tailed Test	One-Tailed Test (Left)	One-Tailed Test (Right)
0.01	±2.58	-2.326	+2.326
0.05	±1.96	**-1.645**	**+1.645**
0.10	**±1.645**	-1.282	+1.282

MCQ

1. What is H_0 usually about?

A) No change

B) New discovery

C) 5% significance

2. Which error occurs when H_0 is wrongly rejected?

A) Type I

B) Type II

C) Type III

MCQ

1. What is H_0 usually about?

A) No change ✅

B) New discovery

C) 5% significance

2. Which error occurs when H_0 is wrongly rejected?

A) Type I ✅

B) Type II

C) Type III

Steps in Hypothesis Testing

1

1. Formulate H_0 and H_1

2

2. Choose α

3

3. Select Test

4

4. Calculate Test Statistic

5

5. Make Decision

Session Summary

In this session, we have discussed about:

- Basics of Hypothesis Testing
- Steps of Hypothesis Testing
- Formulate H_0 and H_1
- Types of Error
- Significance Level
- Types of Statistical Tests

Reference links

- Introduction to Hypothesis Testing
- Frequently asked hypothesis testing interview questions and answers – part 1
- Frequently asked hypothesis testing interview questions and answers – part 2
- Frequently asked hypothesis testing interview questions and answers – part 3

Hypothesis Testing using Statistical Tests

Learning Objectives

By the end of this session, you will be able to:

- Understand when to use which statistical tests
- Apply and practice the various steps of Hypothesis Testing
- Understand the process of one sample and two sample tests.
- Draw inferences and conclusion.
- Understand the concept of degrees of freedom

What is a Hypothesis and Hypothesis Testing?

- An assumption or educated guess about a population parameter.
- Tested using sample data.

- Hypothesis testing is like a detective game where you :

1. start with an assumption (a hypothesis);
2. then collect evidence (data) to decide whether
3. your assumption is likely to be true or not.

- It’s a way of making decisions or inferences about a population based on a sample of data.

Steps in Hypothesis Testing (Recap)

1. Formulate H_0 and H_1

2. Choose α

3. Select Test

4. Calculate Test Statistic

5. Make Decision

Reference

$$\text{Test statistics} = \frac{\textit{Sample statistic} - \textit{Population Parameter}}{\textit{Standard Error}}$$

Commonly used Critical z values

Significance Level (α)	Two-Tailed Test	One-Tailed Test (Left)	One-Tailed Test (Right)
0.01	±2.58	-2.326	+2.326
0.05	±1.96	**-1.645**	**+1.645**
0.10	**±1.645**	-1.282	+1.282

Statistical Test Computations

Testing of Mean – One sample

Testing of Mean – One sample – Large Sample(>=30) and Population SD is known

Test used: Z test (as the sample size is large and population SD is known)

Test statistic:

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Testing of Mean – One sample

Testing of Mean – One sample – Large Sample(>=30) and Population SD is known

Example: Test whether the average height of **18 year old adults in a town** is 158cms with Standard deviation of 2cms. **A sample of 100, 18 year old** was selected and the mean was found to be 159.

Null Hypothesis: There is no significant difference between the sample mean and population mean

Alternate hypothesis: There is significant difference between the sample mean and population mean

Level of Significance: $\alpha = 0.05$

Calculation and Result

Test Statistics: $Z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$

Calculated Value: $Z = \frac{159-158}{2/\sqrt{100}} = 5$ = calculated value

Table value: Z value @ 5% level of significance and two tail test is 1.96 = critical value

Inference: As the calculated value is greater than the table value, Reject Ho. Hence there is significant difference between the sample mean and population mean.

One step ahead: Since the sample mean is greater then the population mean ,

The average height of the population has increased by 1cm. (Inference restricted to this sample study)

Testing of Mean – One sample

Testing of Mean – One sample – **Small Sample(<30)** and Population SD is known

Example: Test whether the average height of 18 year old adult is 158cms with Standard deviation of 2cms. A sample of 25, 18 year old was selected and the mean was found to be 159.

Null Hypothesis: There is no significant difference between the sample mean and population mean

$$H_0: \mu = 158$$

Alternate hypothesis: There is significant difference between the sample mean and population mean

$$H_1: \mu \neq 158$$

Level of Significance: $\alpha = 0.05$

Test Statistics: $Z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$

Calculated Value: $Z = \frac{159-158}{2/\sqrt{25}} = 2.5$

Table value: Z value @ 5% level of significance and two tail test is 1.96

Inference: As the calculated value is greater than the table value, Reject Ho. Hence there is significant difference between the sample mean and population mean.

One step ahead: Since the sample mean is greater then the population mean , The average height of the population has increased by 1cm. (Inference restricted to this sample study)

MCQ

Question: Which test is appropriate for testing a sample mean when the sample size is less than 30 and the population standard deviation is known?

A) F-test

B) T-test

C) Chi-square test

D) Z-test

MCQ

Question: Which test is appropriate for testing a sample mean when the sample size is less than 30 and the population standard deviation is known?

A) F-test

B) T-test

C) Chi-square test

D) Z-test

Answer: D**) Z-test**

Explanation: Even though the sample size is small, if the population standard deviation (SD) is known and the data is approximately normally distributed, a **Z-test** can still be used. However, if the sample size is small and the population SD is unknown, a T-test would be more appropriate.

Testing of Mean – One sample

Testing of Mean – One sample – **Small Sample(<30) and Population SD is unknown**

Test Statistics: t= $\frac{\bar{X}-\mu}{s/\sqrt{n}} \sim$ Student's t distribution with (n-1)df @ 5% level of significance

Where $s = \sqrt{\frac{(X-\bar{X})^2}{n-1}}$

Calculated Value:

Table value: t value with n-1 d.f @ 5% level of significance

Inference: As the calculated value is greater than the table value, Reject Ho. Hence there is significant difference between the sample mean and population mean.

x	X-X-BAR (X – 50)	X-XBAR)2	
80	=80-50 = 30	900	1000/3
30	0	0	333
60	10	100	SQRT(333) = s
30	0	0	

T-table

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768

Degrees of freedom Example

Degrees of Freedom (DoF) represent the number of independent values that can vary in a statistical calculation.

Suppose you have the following 5 exam scores (0 to 100):
x1, x2, x3, x4, and x5 100, 100, 100, 100

You are given that the mean of the scores is 80.

Now suppose the first four scores are allowed to vary freely.

BUT — once 4 of the scores are finalized, the **fifth value is fixed** to maintain the same mean **(80).**

Thus, Degrees of Freedom = n - 1 = 5 - 1 = 4

Testing of Mean – Two Independent sample

Testing of Mean – Two independent sample – Large Sample(>=30) and Population SD is known

Test-statistic:

$$Z = \frac{\bar{x_1} - \bar{x_2}}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

where $\bar{x_1}$ and $\bar{x_2}$ are the sample means, σ_1^2 and σ_2^2 are the population variances,

and n_1 and n_2 are the sample sizes.

Two Independent Sample Z-Test

Problem Statement:

- A **researcher** wants to compare the **average daily calorie intake** of **male and female adults** in a city. Two **independent random samples** are taken:

- **Sample 1 (Males):** n1=40, mean X1_bar =2500, population standard deviation σ1=300
- **Sample 2 (Females):** n2=35, mean X2_bar=2300, population standard deviation σ2=250

- At **5% level of significance**, test whether there is a **significant difference** in the mean calorie intake between males and females.

Step 1: define the hypothesis

- We conduct a **two-sample Z-test** because:
  - **Sample sizes are large** ($n1, n2 \geq 30$)
  - **Population standard deviations ($\sigma 1$, $\sigma 2$) are known**

- **Null Hypothesis (H0)**: There is **no significant difference** between the mean calorie intake of males and females. $H0: \mu 1 = \mu 2$

- **Alternative Hypothesis (H1H_1H1)**: There **is a significant difference** between the mean calorie intake of males and females. $H1: \mu 1 \neq \mu 2$

- This is a **two-tailed test**.

Step 2: Compute the Test Statistic

The **Z-test formula** for two independent samples is:

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Since H_0 assumes $\mu_1 - \mu_2 = 0$, the formula simplifies to:

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Substituting the given values:

$$Z = \frac{2500 - 2300}{\sqrt{\frac{300^2}{40} + \frac{250^2}{35}}}$$

Z comes out to be ?

Step 3: Find the Critical Z-Value

For a **two-tailed test** at **5% significance level (α=0.05)**, we look up the **Z-table**:

Zcritical =±1.96

Step 4: Compare and Make Decision

- **Calculated Z-value = 3.15**
- **Critical Z-value = ±1.96**
- Since Z-calculated > Z critical, **we reject the null hypothesis H0**.

Step 5: Final Inference

- There is **a statistically significant difference** in the mean calorie intake of males and females.

- This suggests that **gender has an influence** on daily calorie consumption.

Session Summary

In this session, we have discussed about:

- Understand when to use which statistical tests
- Apply and practice the various steps of Hypothesis Testing
- Understand the process of one sample and two sample tests.
- Draw inferences and conclusion.
- Understand the concept of degrees of freedom

Reference Links

- Introduction to Hypothesis Testing
- Frequently asked hypothesis testing interview questions and answers – part 1
- Frequently asked hypothesis testing interview questions and answers – part 2
- Frequently asked hypothesis testing interview questions and answers – part 3
- Sampling methods guide
- How to Collect Data Effectively: Methods, Questionnaire Design, and Data Sources Explained
- Probability Distributions in Python
- Step-by-Step Guide to Two-Sample Z-Test for Large Samples (Population SD Known)

A / B testing

Learning Objectives

By the end of this session, you will be able to:

- Understand A/B testing
- Confidence interval role in hypothesis testing
- Concept of p-values in statistical tests

A/B Testing – the popular use case

Scenario
Your company has launched a new website design (Version B) and wants to know if it leads to more conversions compared to the old design (Version A).

The Goal: Use A/B testing to determine if the new design improves the conversion rate.

Sample 1 (old design A): n1=5013, x1=250 conversions
Sample 2 (new design B): n2=4987, x2=325 conversions

At **5% level of significance**, test whether there is a **significant difference** in conversion rates between the two designs.
Let's solve it?

Two proportion test

Two independent sample – Large Sample(>=30)

Test Statistics

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

- The sample proportions are calculated as:

$$\hat{p}_1 = \frac{x_1}{n_1}$$

$$\hat{p}_2 = \frac{x_2}{n_2}$$

- The **pooled proportion p-hat is** calculated as:

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$$

A/B Testing – calculations?

$$\hat{p}_1 = \frac{x_1}{n_1}$$

$$\hat{p}_2 = \frac{x_2}{n_2}$$

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$$

Sample 1 (old design A): n1=5013, x1=250 conversions

Sample 2 (new design B): n2=4987, x2=325 conversions

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

A/B Testing – conclusion

- **Calculated Z-value = -3.29**
- **Critical Z-value = ±1.96**
- Since **|Zcalculated|=3.29>Zcritical=1.96**, we **reject the null hypothesis H0.**
- There is **a statistically significant difference** in the proportion of conversion rate for the two designs.
- This suggests that **website design has an** influence on conversion rates.

Confidence interval and p-values

What is a Confidence Interval in Hypothesis Testing?

A **confidence interval (CI)** is a range of values that is likely to contain the **true population parameter** (like the mean or proportion) with a certain level of confidence (typically 95%).

Instead of saying, “The average weight is 70 kg,” we say:
👉 “We are 95% confident the average weight lies between 67 and 73 kg.”

It provides more information than a point estimate and helps in decision-making — if your hypothesized value (e.g., H_0 mean) falls **outside** the interval, it’s likely **not true**.

Formula

Z Confidence Interval (when population standard deviation is known)

$$CI = \bar{x} \pm Z\left(\frac{\sigma}{\sqrt{n}}\right)$$

Where:

- $\bar{x}$ = sample mean
- Z = Z-score for desired confidence level (e.g., 1.96 for 95%)
- σ = population standard deviation
- n = sample size

t Confidence Interval (when population standard deviation is unknown)

$$CI = \bar{x} \pm t\left(\frac{s}{\sqrt{n}}\right)$$

Where:

- t = t-value based on degrees of freedom (df = n−1)
- s = sample standard deviation

Solved Example – Z-Test CI

Problem:
A sample of 100 students has an average score of 75. The population standard deviation is 10. Find the **95% confidence interval** for the true mean.

$$CI = \bar{x} \pm Z\left(\frac{\sigma}{\sqrt{n}}\right)$$

Solved Example – Z-Test CI

Problem:
A sample of 100 students has an average score of 75. The population standard deviation is 10. Find the **95% confidence interval** for the true mean.

Solution:

$$\bar{x} = 75, \quad \sigma = 10, \quad n = 100, \quad Z_{0.025} = 1.96$$

$$CI = 75 \pm 1.96 \times \left(\frac{10}{\sqrt{100}} \right) = 75 \pm 1.96 \times 1 = 75 \pm 1.96$$

95% CI = (73.04, 76.96)

Solved Example – t-Test CI

Problem: A sample of 16 patients has an average blood pressure of 120. The sample standard deviation is 8. Find the **95% confidence interval** assuming a t-distribution.

$$CI = \bar{x} \pm t\left(\frac{s}{\sqrt{n}}\right)$$

Solution:

$$\bar{x} = 120, \quad s = 8, \quad n = 16, \quad df = 15, \quad t_{0.025,15} \approx 2.131$$

$$CI = 120 \pm 2.131 \times \left(\frac{8}{\sqrt{16}}\right) = 120 \pm 2.131 \times 2 = 120 \pm 4.262$$

95% CI = (115.74, 124.26)

p-values

What is a p-value in Hypothesis Testing?

The **p-value** (or probability value) tells us how likely we are to observe the data we have **if the null hypothesis (H_0) were true**.

In simple words:

It measures the **evidence against H_0**.

- **Small p-value** → strong evidence against H_0
- **Large p-value** → weak evidence against

Example:

A company claims their average delivery time is **30 minutes**. You collect data and find the average is **34 minutes**. After running a hypothesis test, your **p-value = 0.01**. What does it mean?

Interpretation:

Ho: average delivery time = 30 mins

If H_0 (delivery time is 30 mins) were true, there's only a **1% chance** of getting a result like 34 mins or more extreme.

Now, Since **p = 0.01 < 0.05**, we **reject H_0**.

We conclude that the actual delivery time is **significantly different** from 30 mins.

MCQ

Q1. $p=0.03 \rightarrow ?$

A) Accept H_0

B) Reject H_0

Q2. CI=(48,52), $\mu_0=53 \rightarrow ?$

A) Reject H_0 when your population or sample mean is outside the CI, reject the null hypothesi

B) Accept H_0

Q3) $p=0.07$, $\alpha=0.05 \rightarrow ?$

A) Reject H_0

B) Fail to reject H_0

MCQ

Q1. $p=0.03 \rightarrow ?$

A) Accept H_o

B) Reject H_o ✅

Q2. $CI=(48,52)$, $\mu_o=53 \rightarrow ?$

A) Reject H_o ✅

B) Accept H_o

Q3) $p=0.07$, $\alpha=0.05 \rightarrow ?$

A) Reject H_o

B) Fail to reject H_o ✅

Session Summary

In this session, we have discussed about:

- Understand A/B testing
- Confidence interval role in hypothesis testing
- Concept of p-values in statistical tests

Reference Links

- How to perform A/B Testing in Python
- Introduction to Hypothesis Testing
- Frequently asked hypothesis testing interview questions and answers – part 1
- Frequently asked hypothesis testing interview questions and answers – part 2
- Frequently asked hypothesis testing interview questions and answers – part 3
- Sampling methods guide
- How to Collect Data Effectively: Methods, Questionnaire Design, and Data Sources Explained
- Probability Distributions in Python
- Step-by-Step Guide to Two-Sample Z-Test for Large Samples (Population SD Known)